A production bias model of the Constant Rate Effect

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Introduction

Aims of this talk:

- Outline some problems with the Constant Rate Hypothesis (CRH) as standardly described
- Provide a mechanism for Constant Rate Effects using a modified version of the learner in Yang (2000)
- Show that this model fits the data as well as or better than the traditional approach
- Illustrate an interesting consequence: when biases are in play, the grammar which can parse the most sentence types does not automatically win!

1

The Constant Rate Hypothesis and Constant Rate Effects

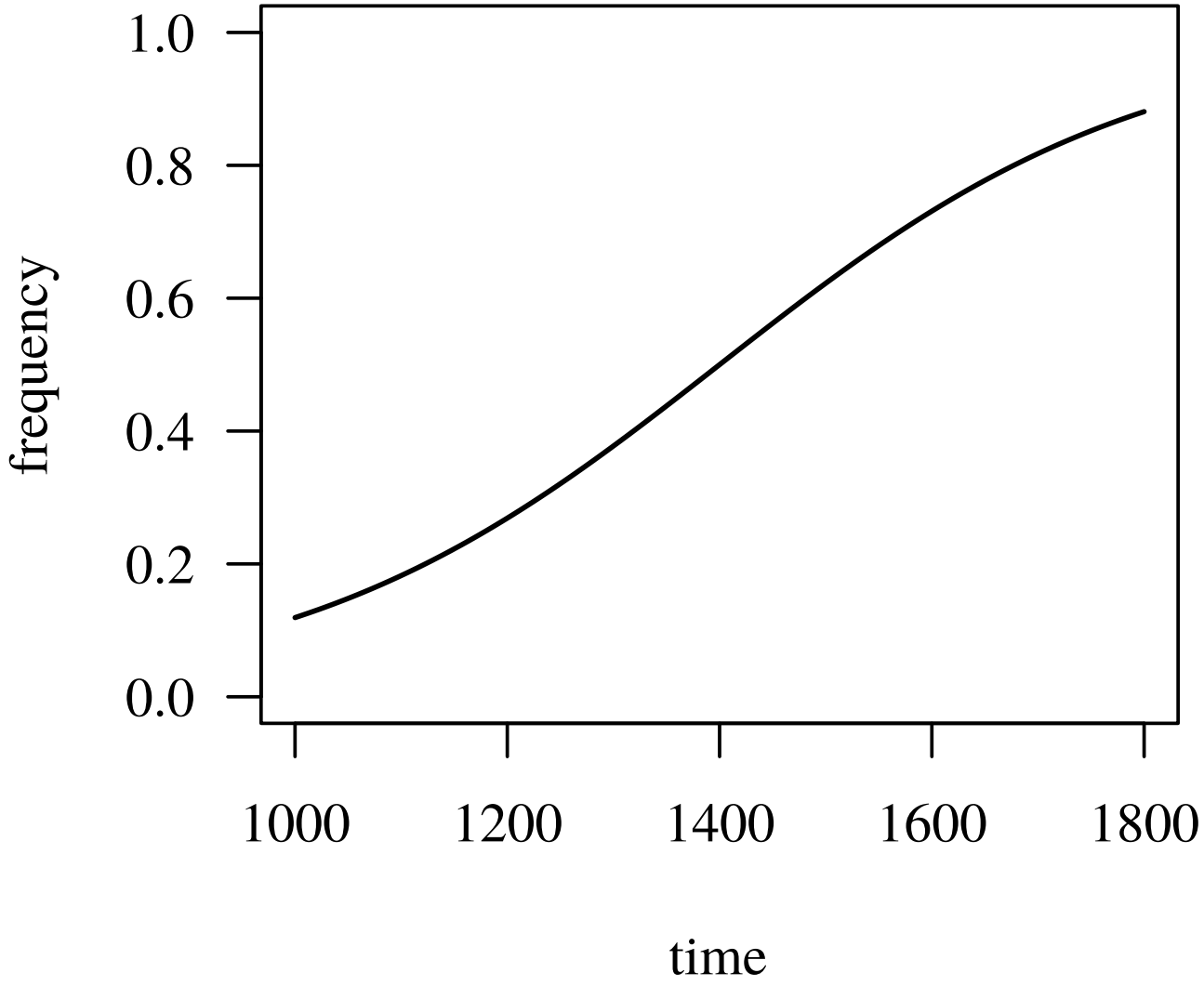
S-curves and the logistic equation

- Altmann et al. (1983), Kroch (1989): change is governed by the logistic equation.

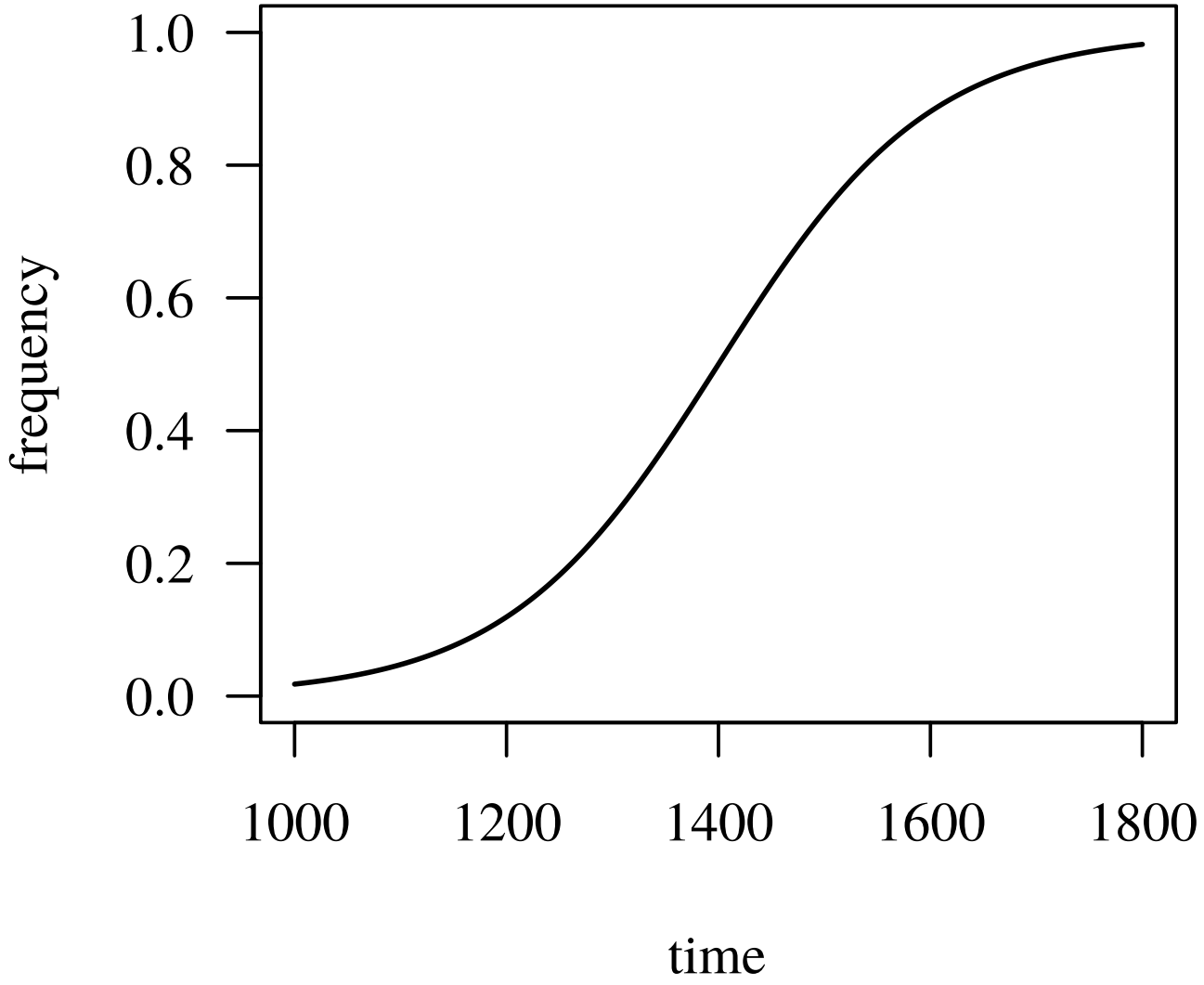
$$p(t) = \frac{1}{1 + e^{-s(t-k)}} \tag{1}$$

- The s parameter gives the overall steepness of the curve.
- The k parameter tells us where, in time, the curve is situated.

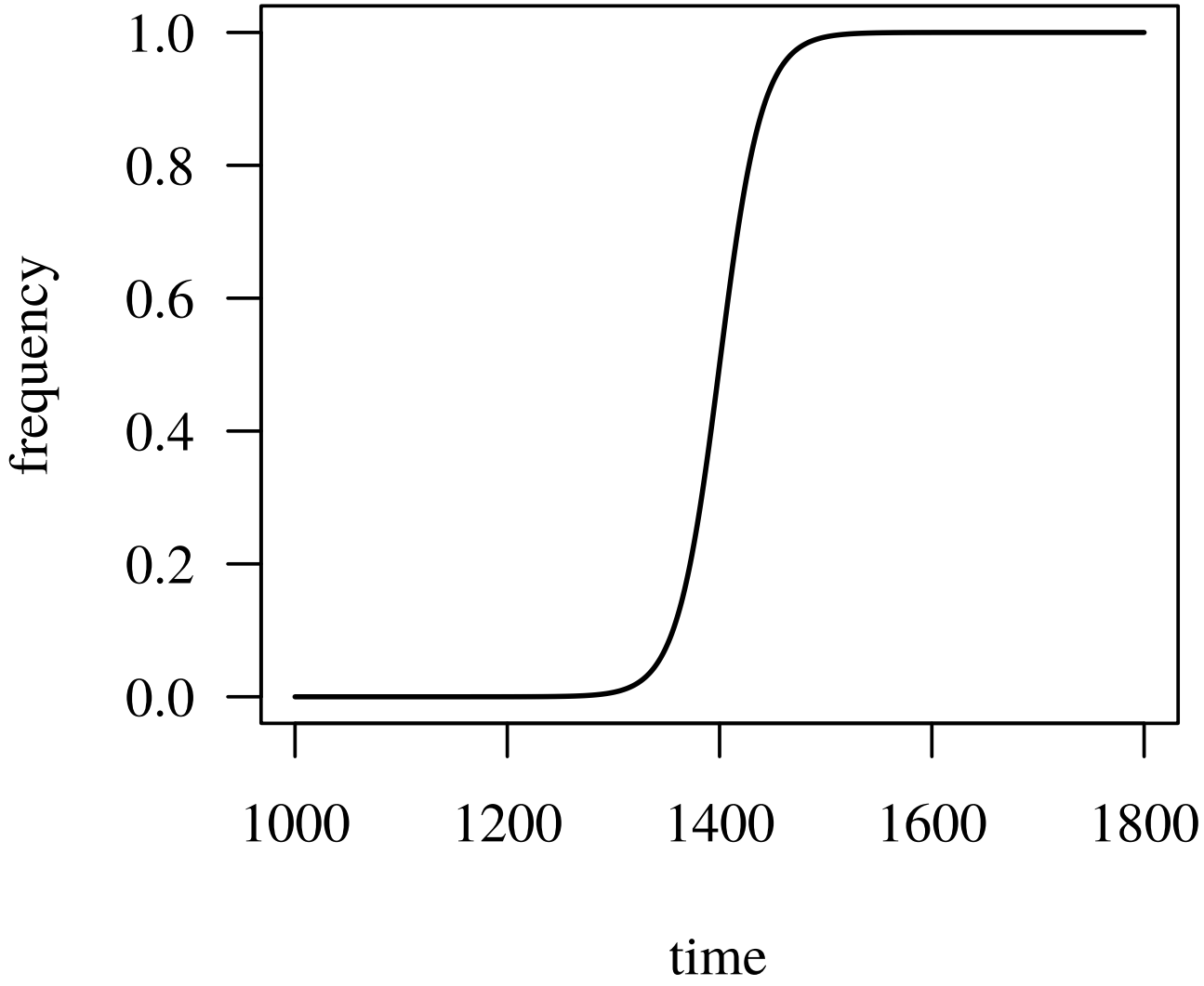
Varying the s parameter



Varying the s parameter



Varying the s parameter



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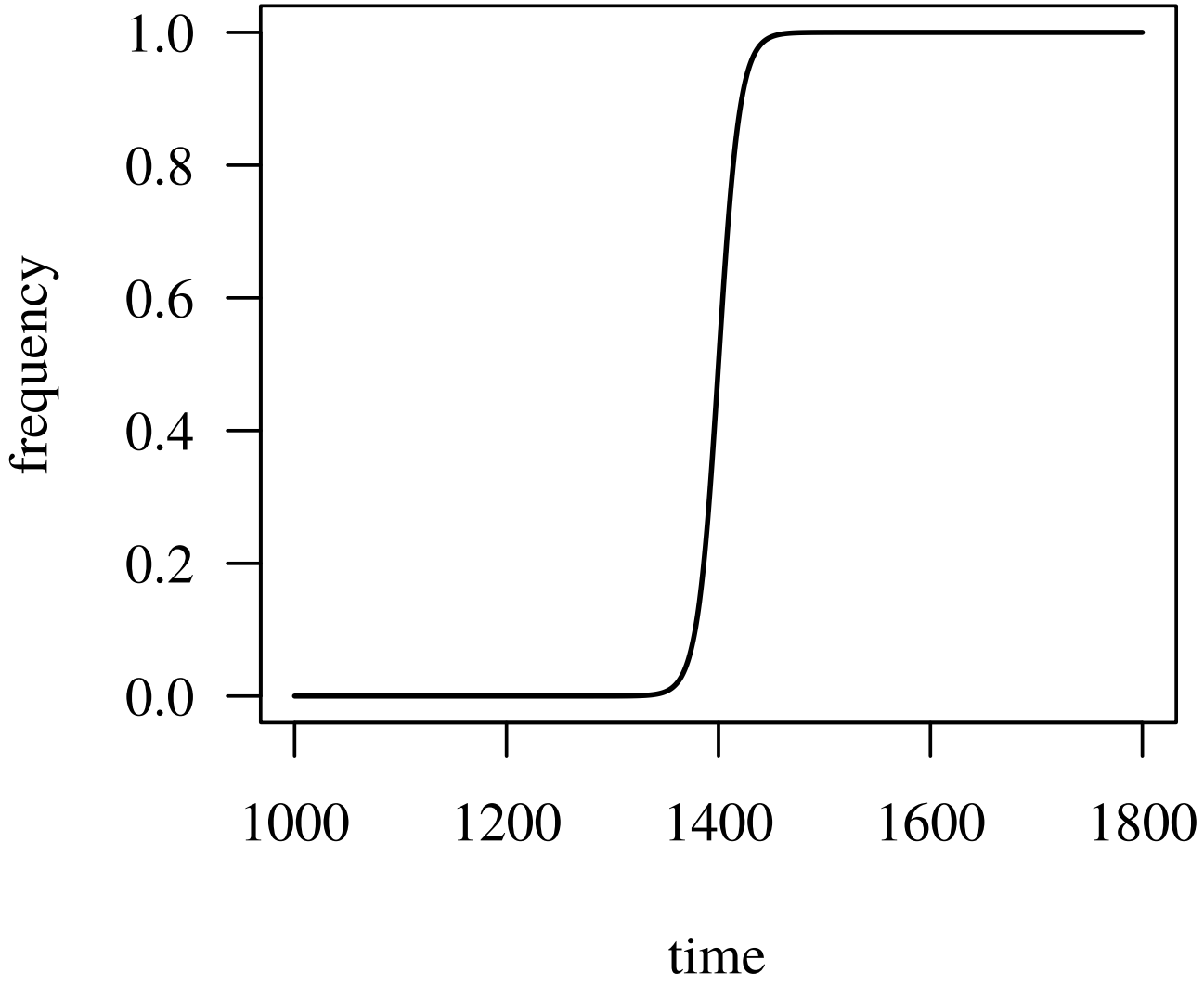
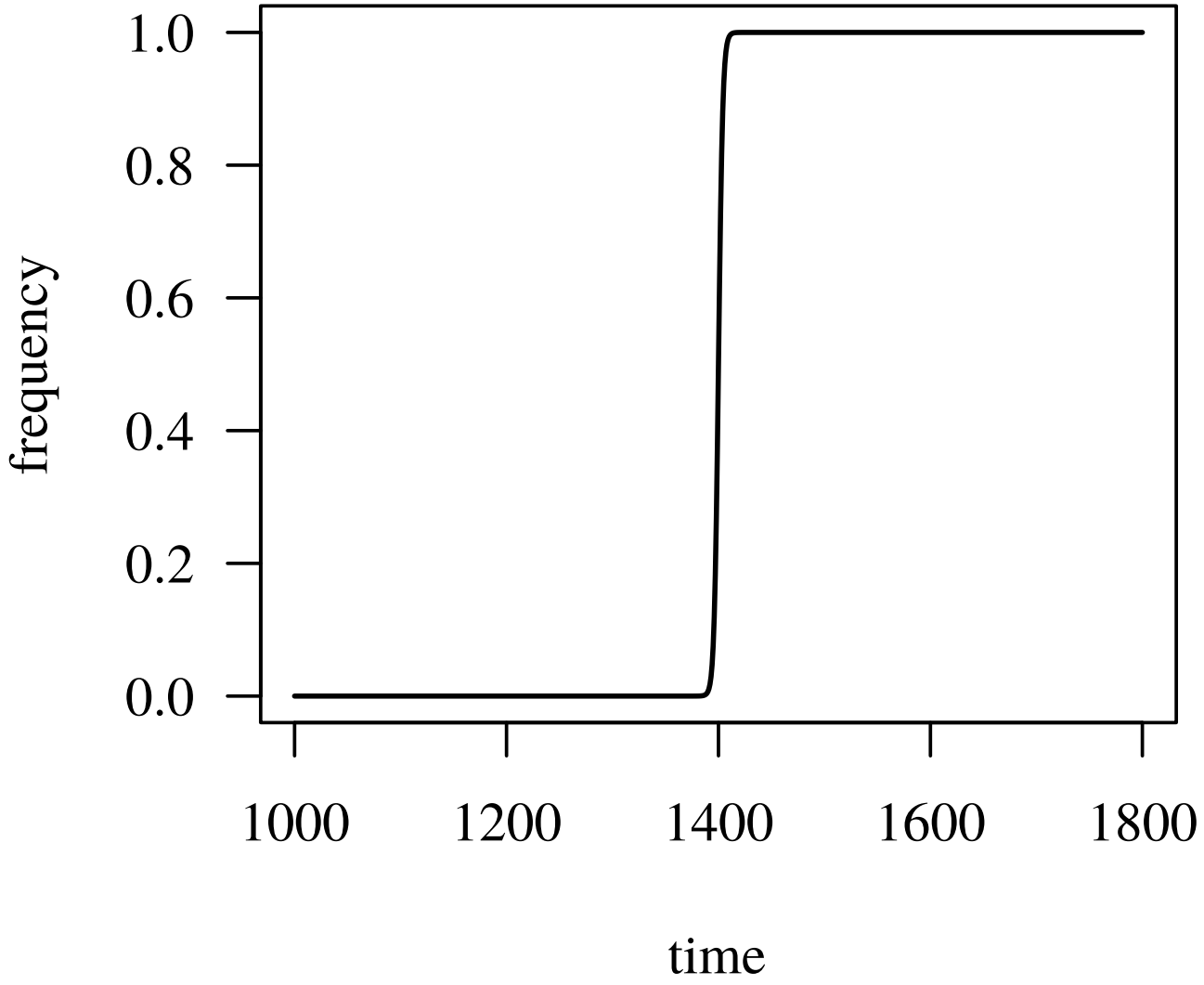
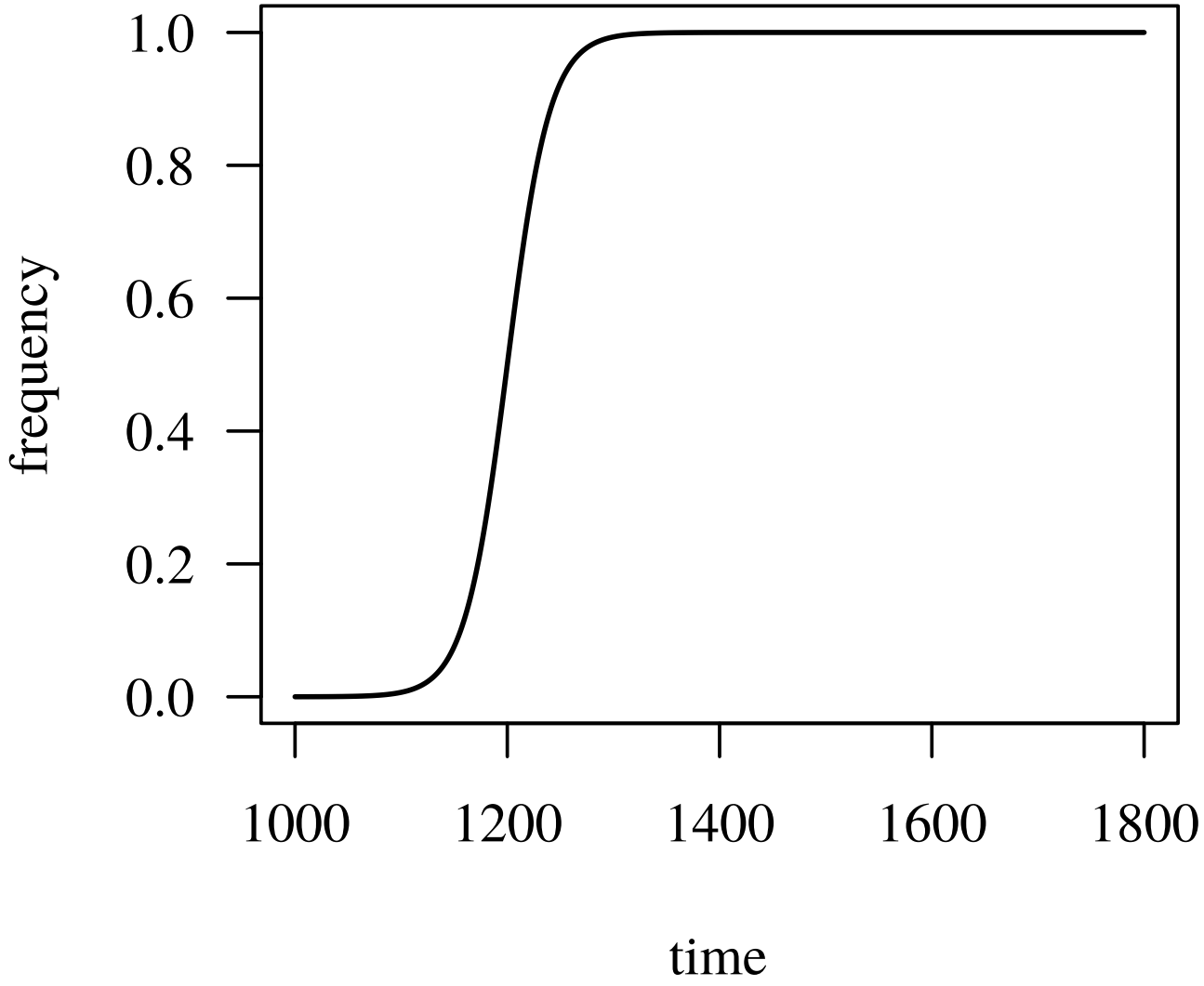


Varying the s parameter



Varying the k parameter



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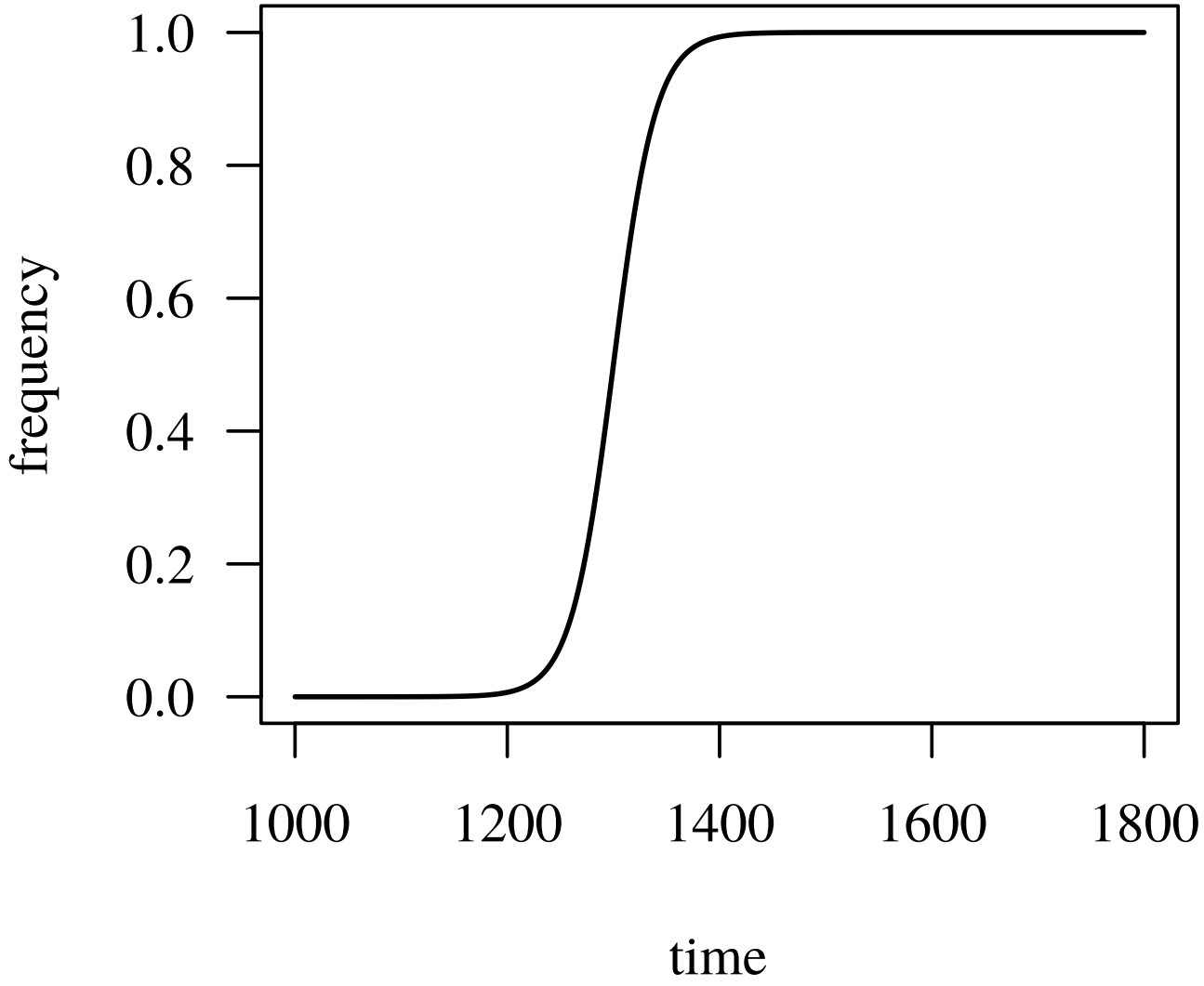
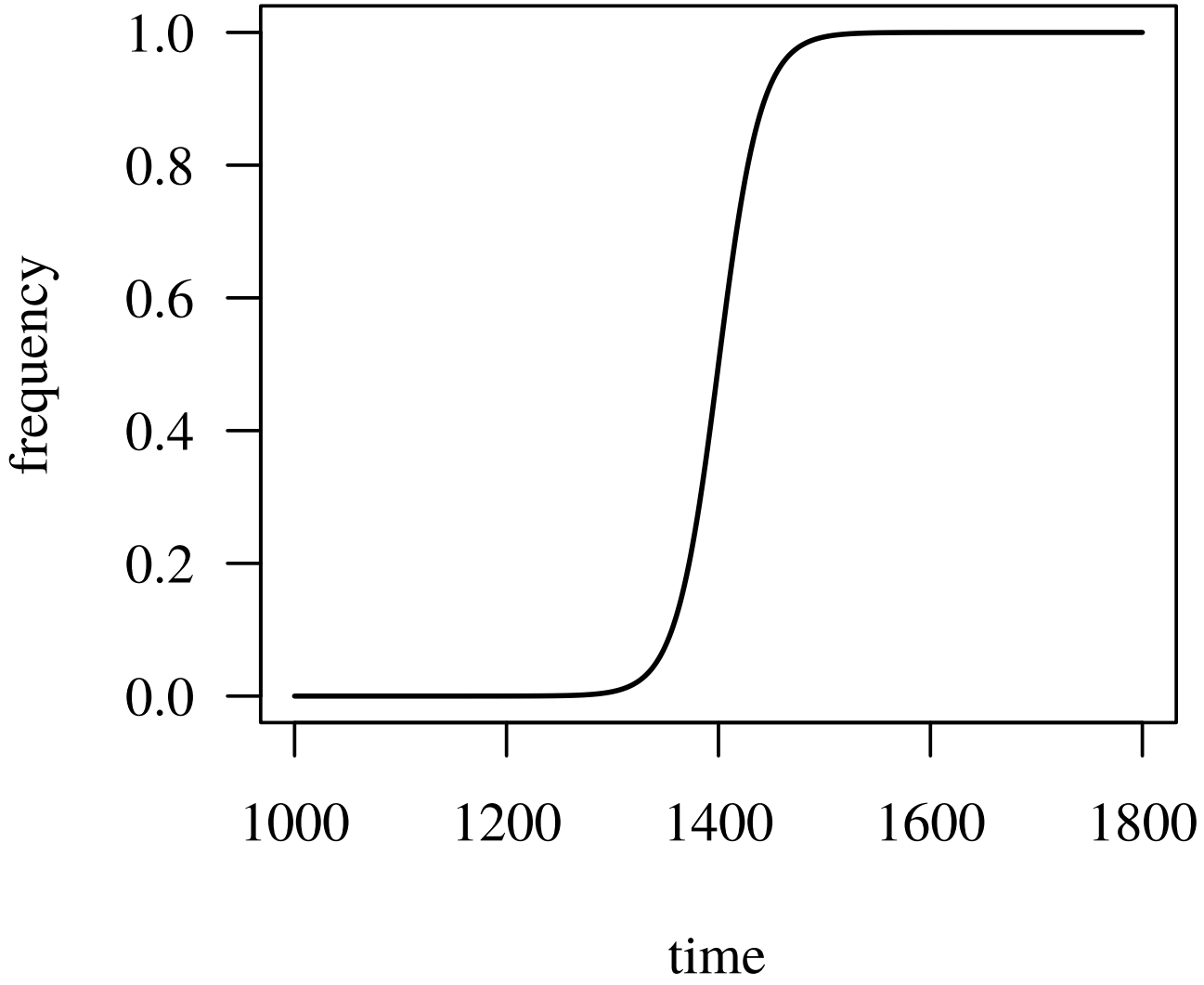


Varying the k parameter



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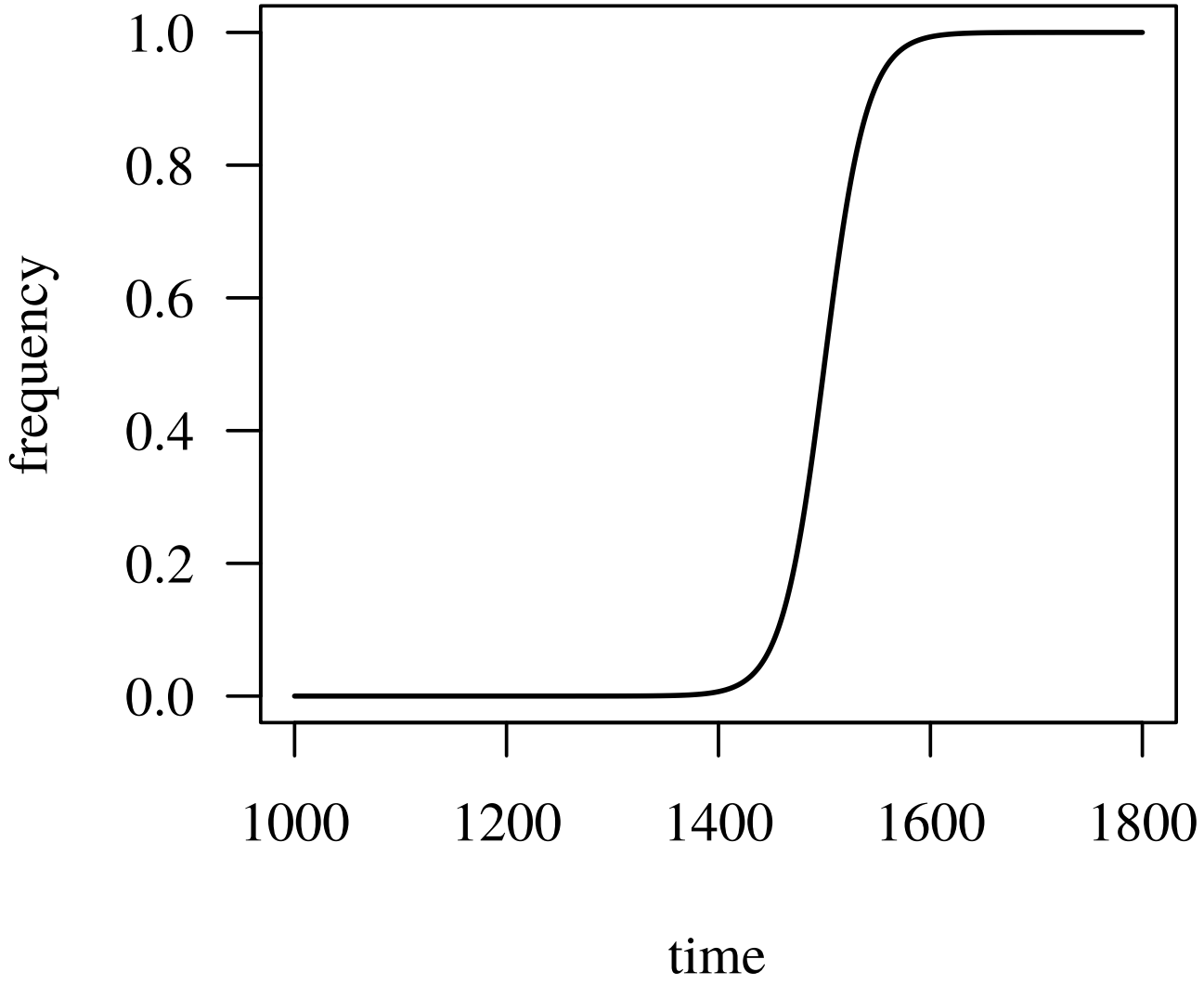
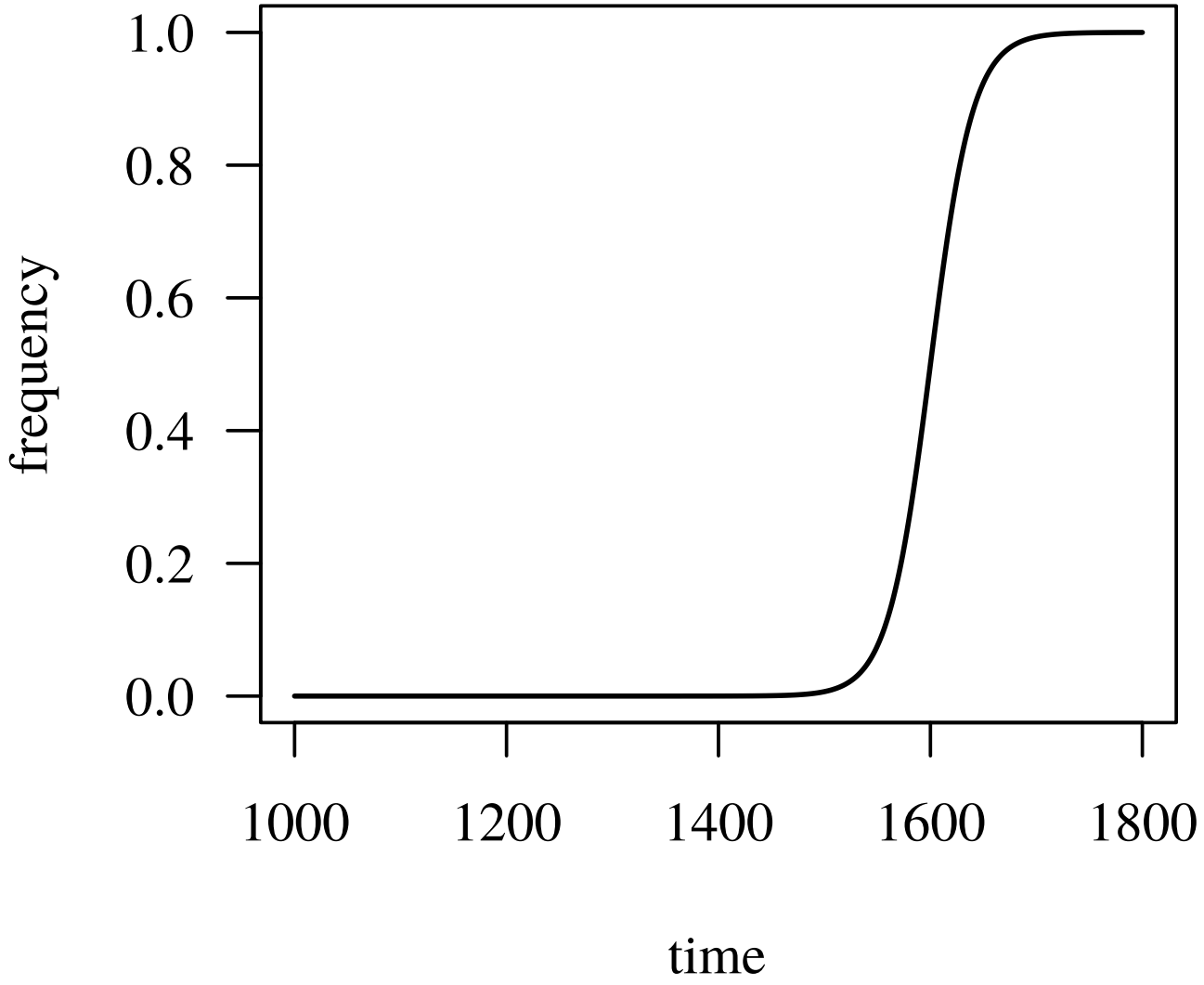


Varying the k parameter

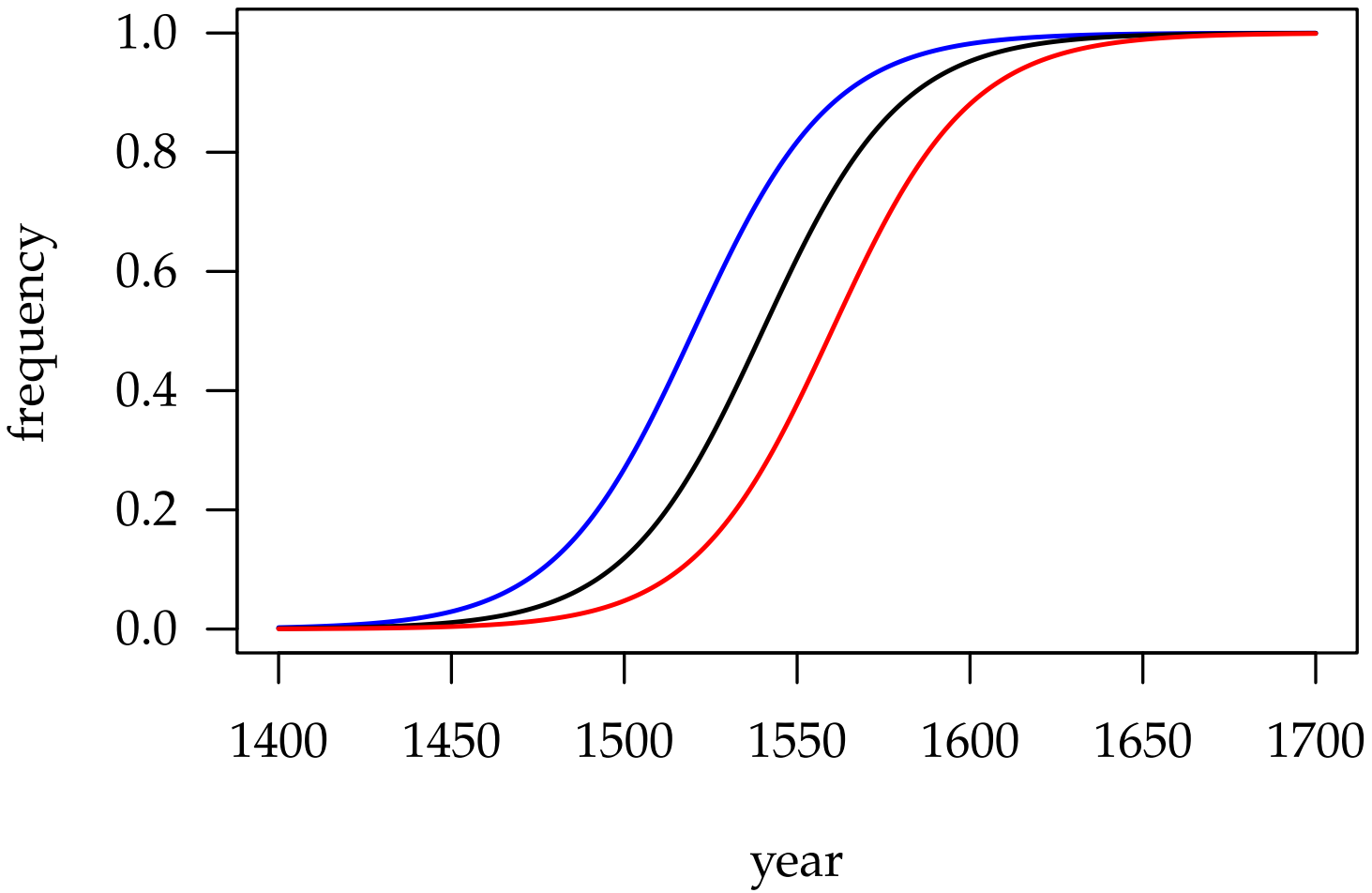


Constant Rate Hypothesis (CRH)

"[W]hen one grammatical option replaces another with which it is in competition across a set of linguistic contexts, the rate of replacement, properly measured, is the same in all of them." (Kroch, 1989, 200)

- In terms of the logistic equation: the s values are the same, but the k values may differ.
- Since Kroch (1989), many empirical studies have found this sort of situation (henceforth Constant Rate Effects, CREs).
- Interpretation (Kroch 1989): a single syntactic parameter governs all contexts. Some contexts may favour the new form, others disfavour it, but those (dis)favouring effects are **CONSTANT**.
- Relies on individual speakers having access to multiple, probabilistically-weighted **COMPETING GRAMMARS**.

Different linguistic contexts

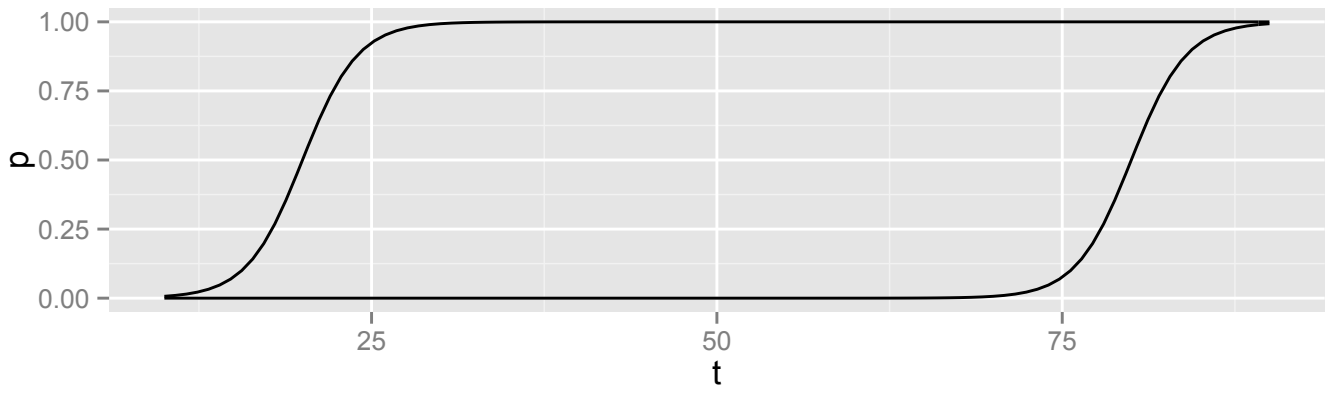


Problem 1: no mechanism

- Kroch (1989, 204) adopts the logistic equation because of its use in biology, but admits that it does not follow from anything.
- Since then, Yang (2000) has derived the logistic equation from his model of syntactic acquisition and change (on which more soon).
- But no mechanism has so far been presented **TO DERIVE THE CRH**: how exactly are the contexts tied to an underlying change?

Problem 2: time separations

- For a CRE, the s values must not differ significantly. But what about large differences in k?



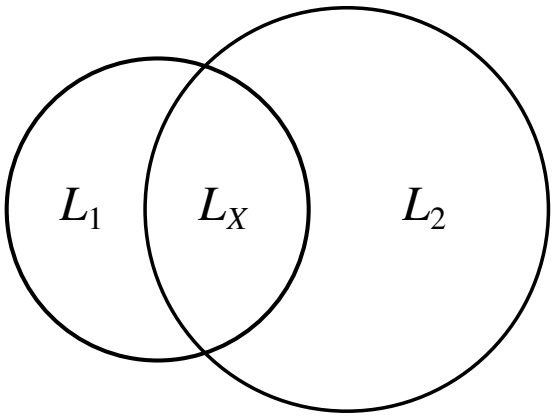
- If the change only gets off the ground in one context 50 years after it has run to completion in another context (or 500, or 5,000,000), how plausible is it that we are dealing with reflexes of a single parametric change? (Think of the acquirer!)
- But this would still qualify as a CRE.

2

The model

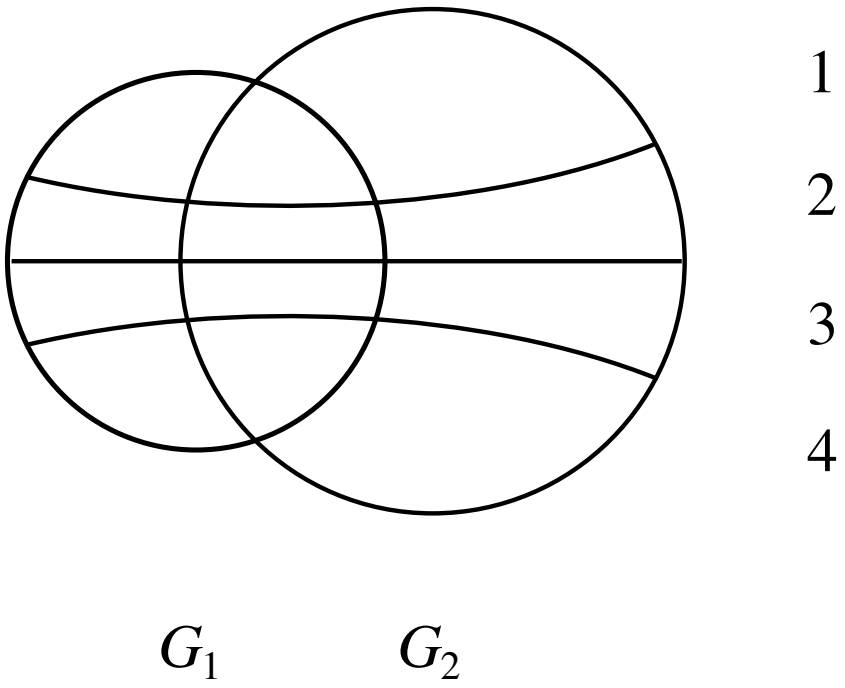
Change

- We assume Yang (2000) here
- Two grammars G_1 and G_2 compete: speakers use G_1 with probability p and G_2 with probability $q = 1 - p$
- The **ADVANTAGE** of G_1, α, is the proportion of sentences G_2 does not parse but G_1 does
- Vice versa for G_2, β
- The ratio α/β determines the dynamics
- Assuming infinite, well-mixing, non-overlapping generations and reliable learners, change is logistic



Production biases

- Now assume that the PLD is partitioned by M contexts $C_1, \ldots, C_M$, illustrated here with $M = 4$:



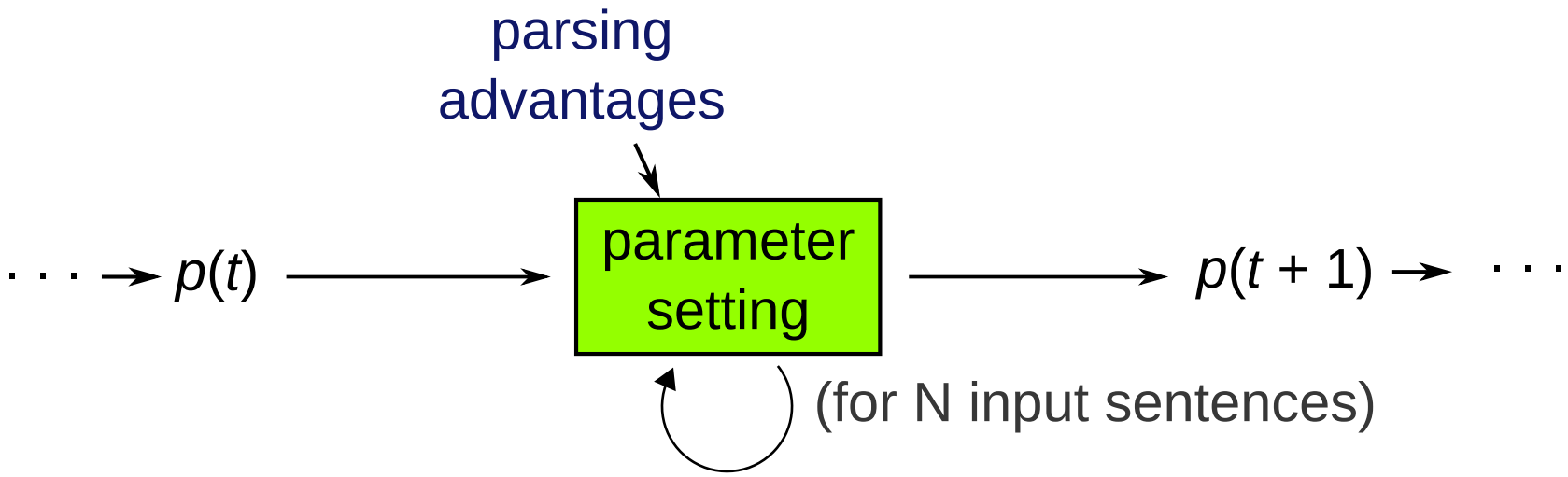
- Key idea: CREs occur if different **PRODUCTION BIASES** affect different contexts

Production biases

- New terminology:
  - $p_i = p_i(t)$ = prob. of G_1 in the ith context
  - $q_i = q_i(t)$ = prob. of G_2 in the ith context
- The general form of biasing is

$$\begin{cases} p_i = p + F(b_i, p, q) \\ q_i = q + G(b_i, p, q) \end{cases} \tag{2}$$

  for context-sensitive production bias b_i and some (yet unknown) functions F, G
- p (q) is the baseline probability of G_1 (G_2) set during learning


parsing
advantages
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p(t)
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p(t + 1)
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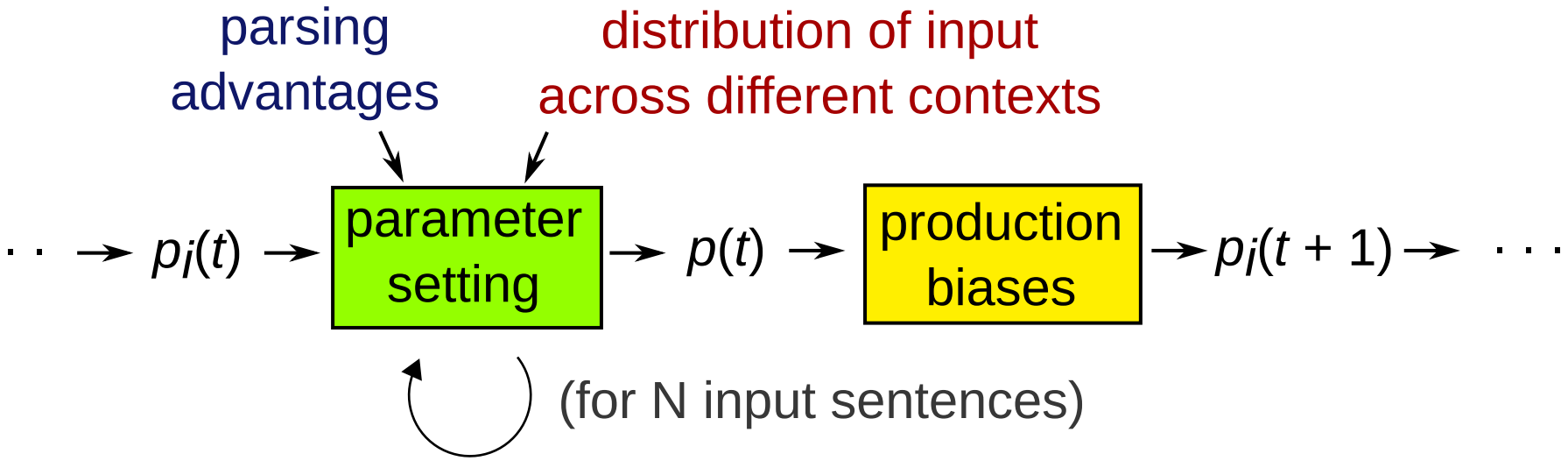
(for N input sentences)


parsing
advantages
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p_i(t)
parameter
setting
p(t)
production
biases
p_i(t + 1)
· · ·
(for N input sentences)

Production biases

- Two requirements:
  (R1) $p_i + q_i = 1$ (as G_1 and G_2 are the only grammars)
  (R2) $0 \leq p_i, q_i \leq 1$ (as these need to be probabilities)
- ICBS: (R1) and (R2) are satisfied only if

$$\begin{cases} F & = & -G \\ |F|, |G| & \leq & \min\{p, q\} \end{cases} \tag{3}$$

- The most parsimonious choice is to put

$$\begin{cases} F = -b_i pq \\ G = b_i pq \end{cases} \tag{4}$$

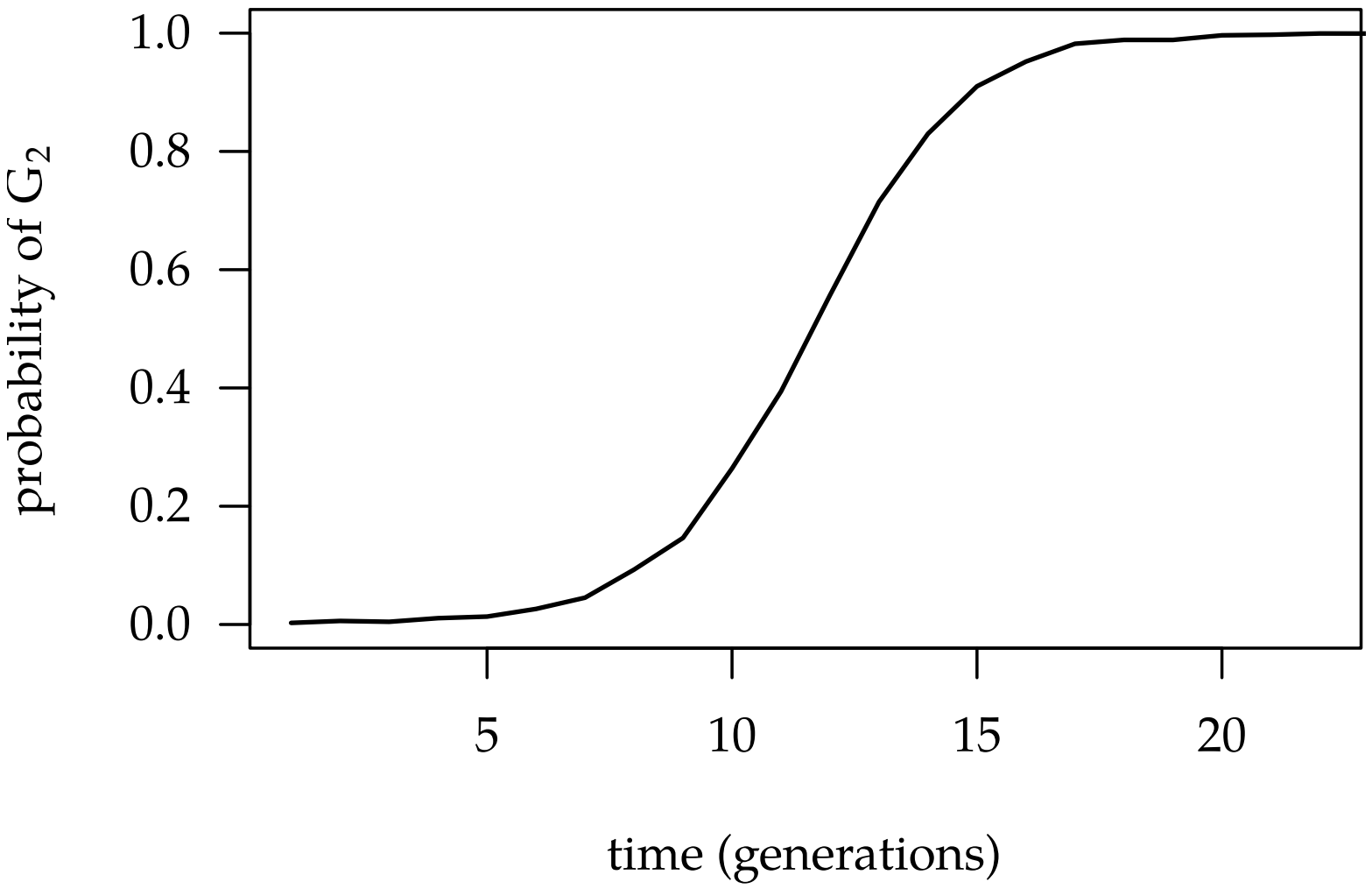
with $-1 \leq b_i \leq 1$.

Production biases

- Thus, the production biasing works as

$$\begin{cases} p_i = p - b_i pq \\ q_i = q + b_i pq \end{cases} \tag{5}$$

  where b_i is a **CONSTANT** production bias (does not change over time)
- if $b_i > 0$, G_2 is favoured in context i
- if $b_i < 0$, G_1 is favoured in context i
- Dynamics of system comes to depend on a combination of grammar advantages, production biases, and how many sentences fall in each context


1.0
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probability of G_2
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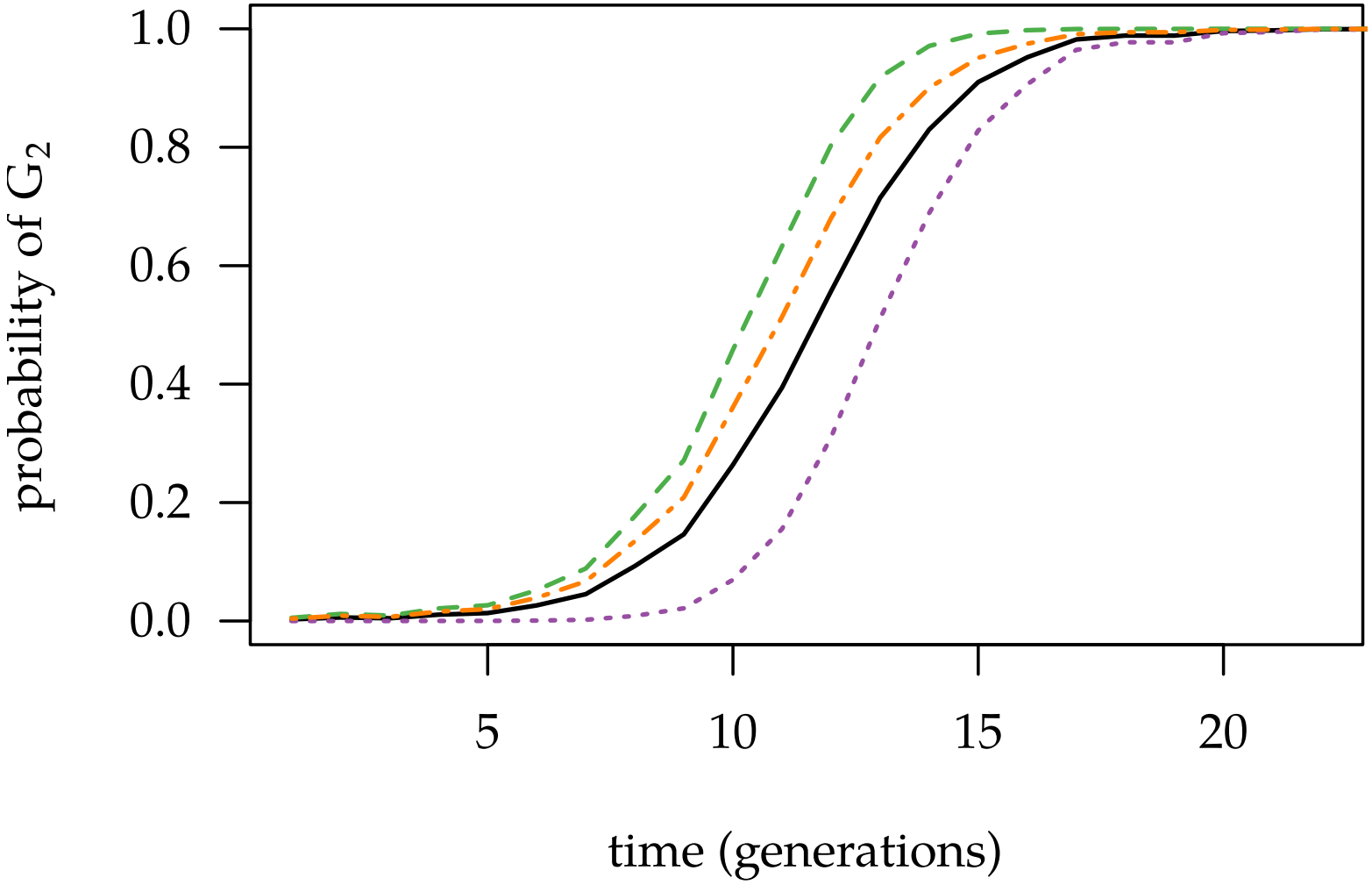
20
time (generations)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probability of G_2
5
10
15
20
time (generations)

Production biases

- System evolves according to

$$q(t+1) = \left(1 + \frac{\sum_{i=1}^{M} \alpha\lambda_i(p(t) - b_i p(t)q(t))}{\sum_{i=1}^{M} \beta\lambda_i(q(t) + b_i p(t)q(t))}\right)^{-1} \tag{6}$$

  where λ_i = proportion of PLD in ith context
- A feedback loop between advantage and bias – interesting
- Surprising(?): this is still logistic almost everywhere
- Will return to the cases where it isn't…

The Time Separation Theorem

- Logisticity means we can derive each context curve p_i from an underlying logistic p:

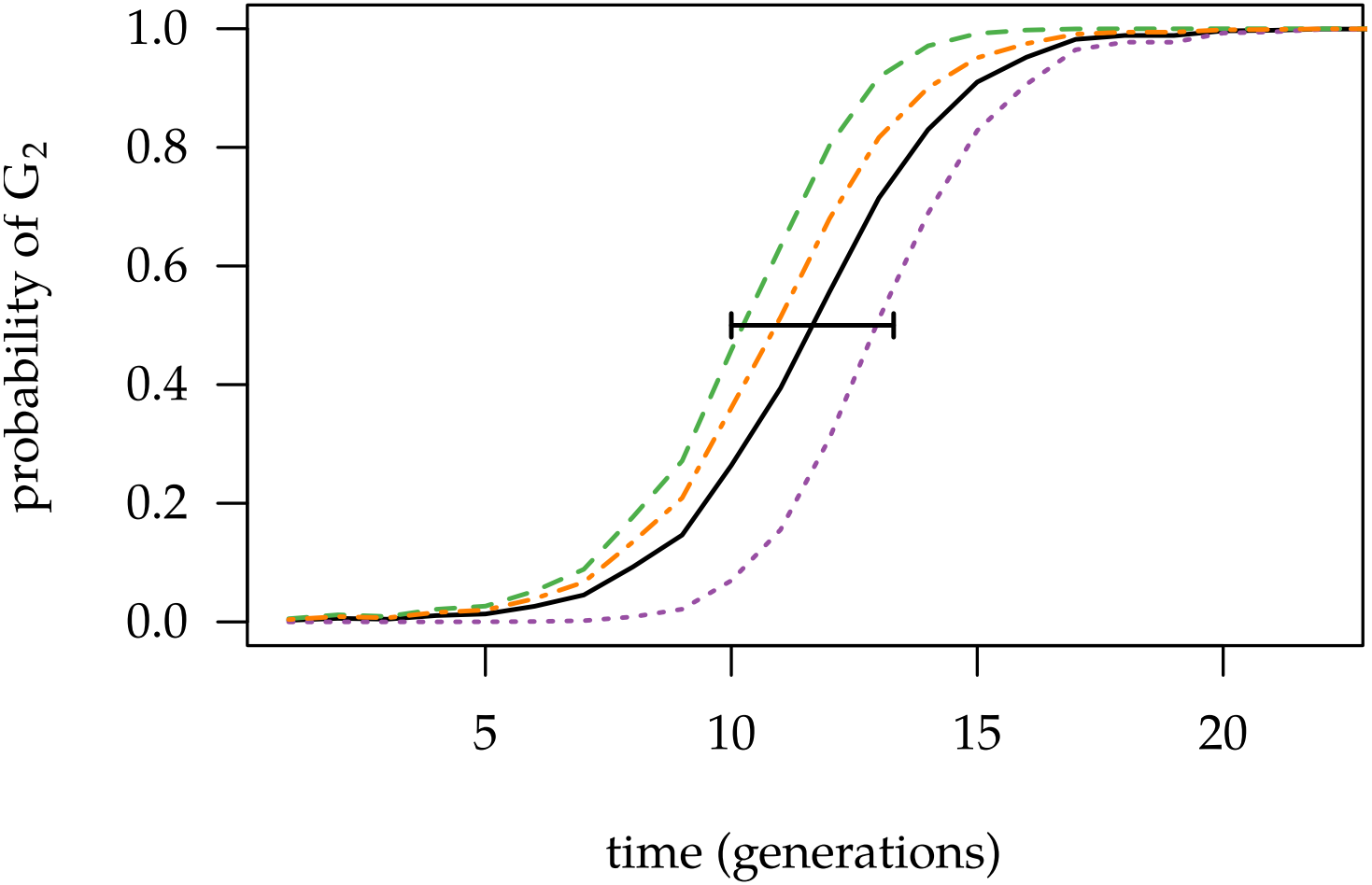
$$p_i = p + b_i p(1 - p) \tag{7}$$

  where $-1 \leq b_i \leq 1$.
- Importantly, this implies:

Theorem (The Time Separation Theorem)

For an underlying change $p(t)$ logistic with slope s, the maximal time separation between two contexts at the tipping points is

$$\Delta_{\max}(s) = \frac{2}{|s|} \log\left(\frac{1}{\sqrt{2} - 1}\right) \approx 1.76 \cdot \frac{1}{|s|}. \tag{8}$$


1.0
0.8
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0.4
0.2
0.0
probability of G_2
5
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time (generations)

Production biases

- NB the above production biasing mechanism can in principle be applied to **ANY** underlying change, not necessarily Yangian, not necessarily logistic, not necessarily even monotonic
- Here, we keep assuming Yang for concreteness

3

Evaluating the model: empirical fit

Curve fitting procedure

- We assume an underlyingly logistic change $p_{s,k}$ with slope s and intercept k, i.e

$$p_{s,k} = \frac{1}{1 + e^{-s(t-k)}} \tag{9}$$

- Define reasonable ranges of variation for s and k
- Loop over this range, fitting a curve of the form

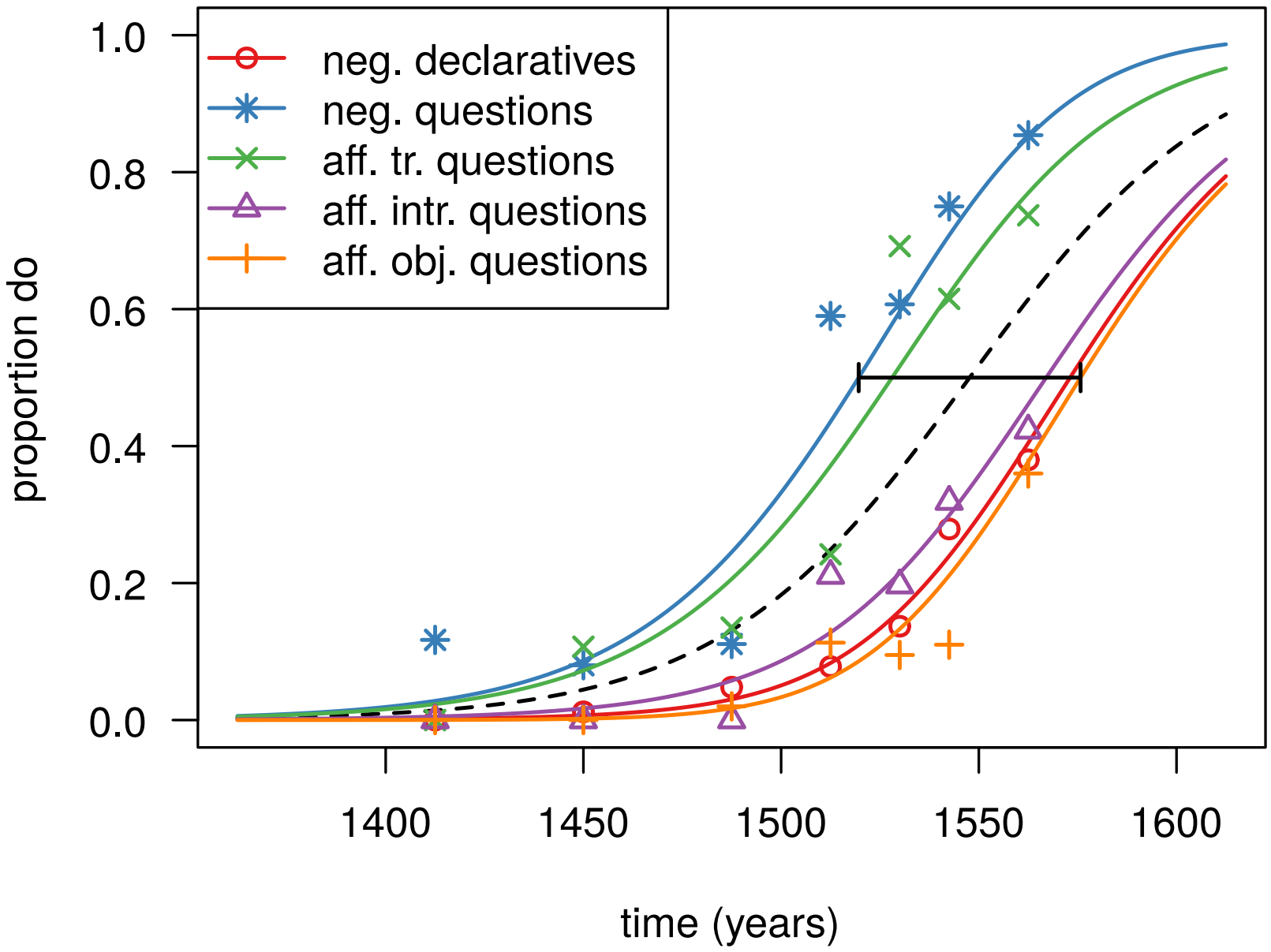
$$p_i = p_{s,k} + b_i p_{s,k}(1 - p_{s,k}) \tag{10}$$

  for each context C_i
- The best-fitting b_i is found by nonlinear optimization (gradient descent)
- Return best combination of $(s, k, b_1, \ldots, b_M)$

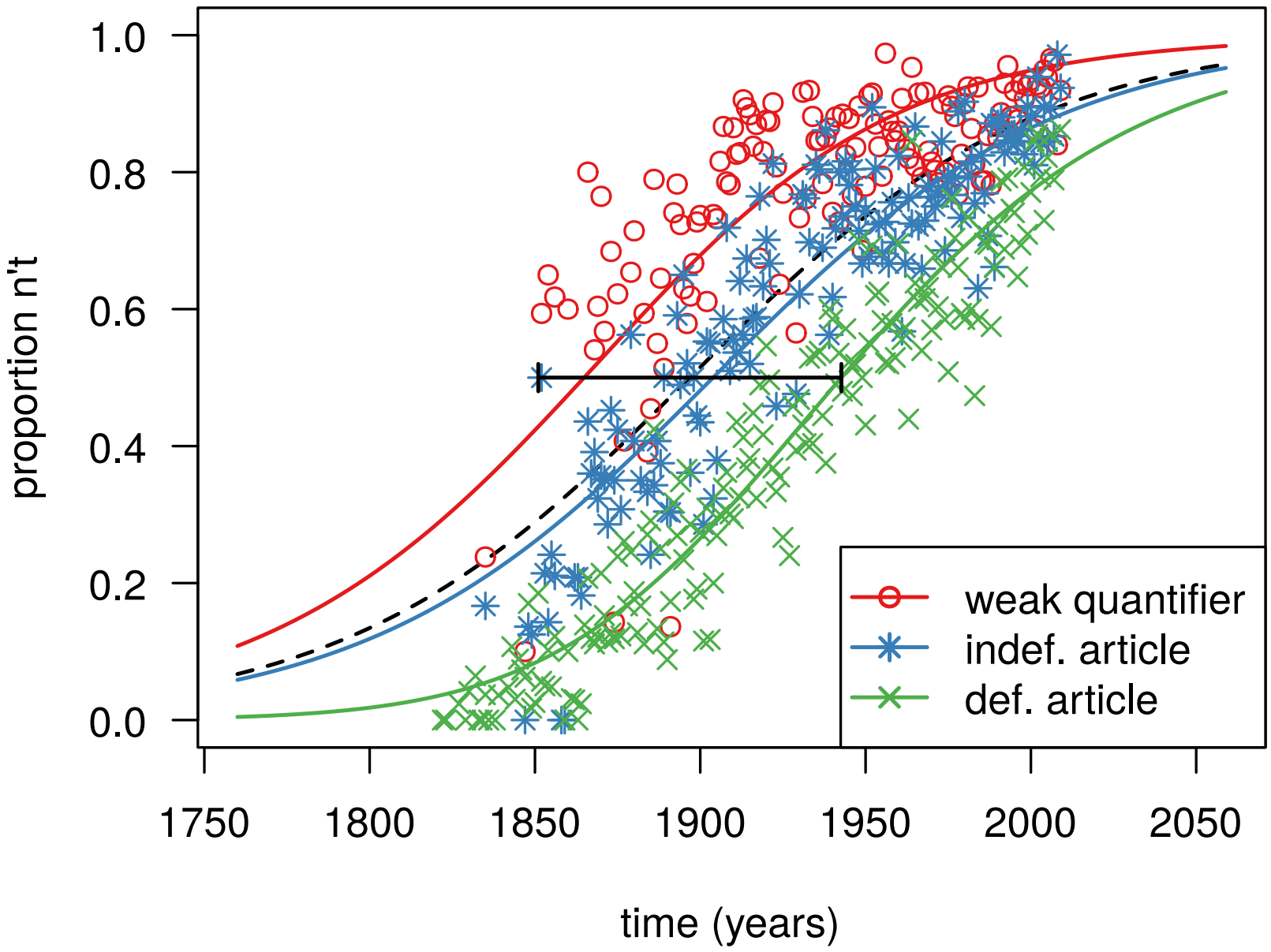
Curve fitting procedure

- Test model fit in four ways:
  - vis-à-vis Time Separation Theorem
  - residual error of fit
  - against a "control model" that fits logistic curves varying in their intercepts (k) but having the same slope (s)
  - see how model fares on a "pseudo-CRE"
- Four case studies:
  - *do*-support in English (Kroch, 1989)
  - not/n't with *have* and *be* in AmEng (Zimmermann, p.c.)
  - final fortition in Early New High German (Fruehwald et al., 2009)
  - Manchester /t/-glottalling (Bermúdez-Otero et al., 2016)

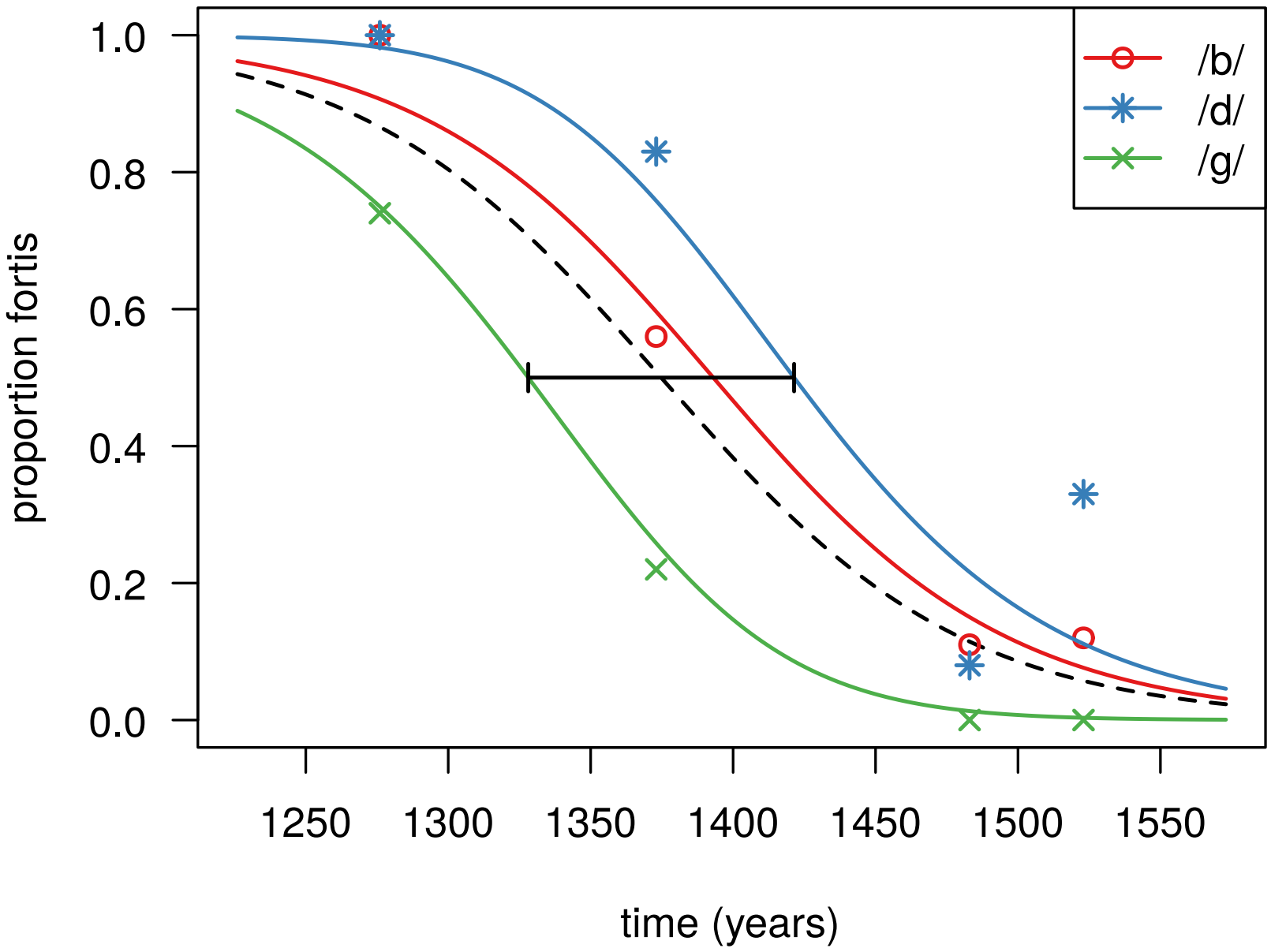
do-support in English (Kroch, 1989)



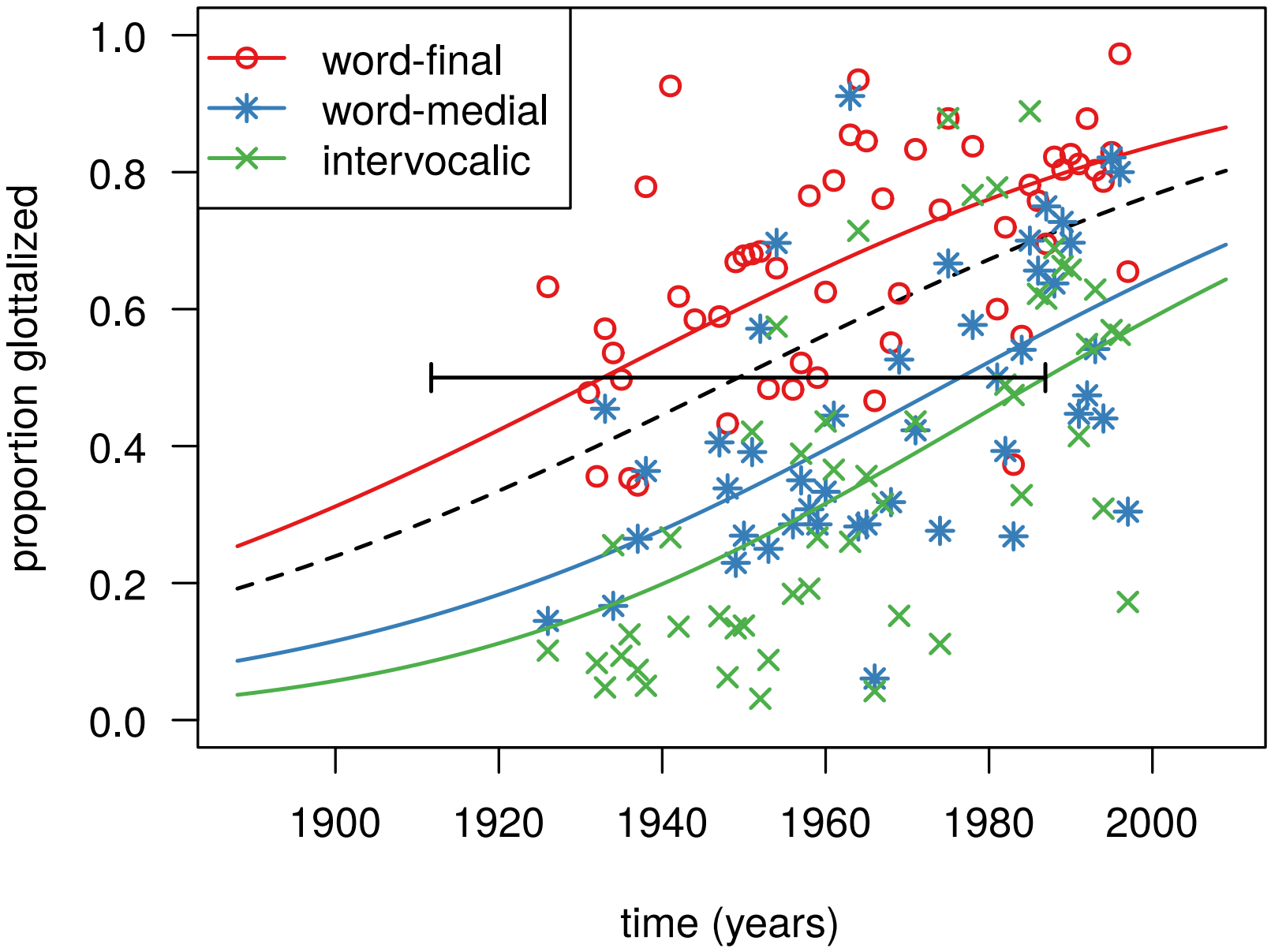
Negative cliticization in AmEng (Zimmermann, p.c.)



German final fortition (Fruehwald et al., 2009)



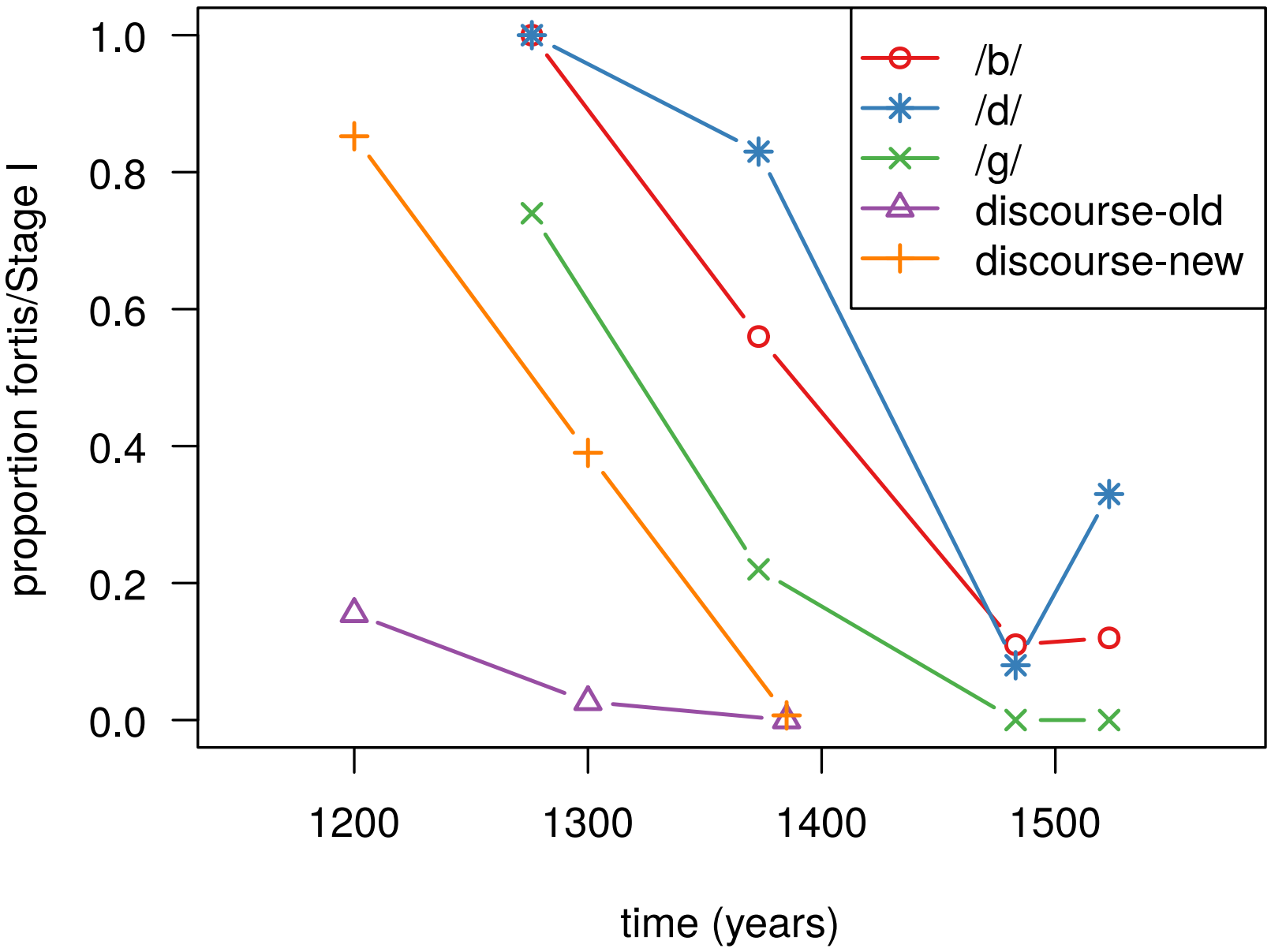
Manchester /t/ (Bermúdez-Otero et al., 2016)



Pseudo-CRE

- Test model on a **PSEUDO-CRE**: a case where a number of variables change at a similar rate but cannot be considered reflexes of one underlying change
- Here, we put together the German final fortition case study (Fruehwald et al., 2009) with Jespersen's Cycle in English, where Wallage (2013) reports a CRE between discourse-old and discourse-new propositions
- If probed using the standard operationalization, both changes suggest a slope of $s \approx -0.02$

Pseudo-CRE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proportion fortis/Stage I
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time (years)
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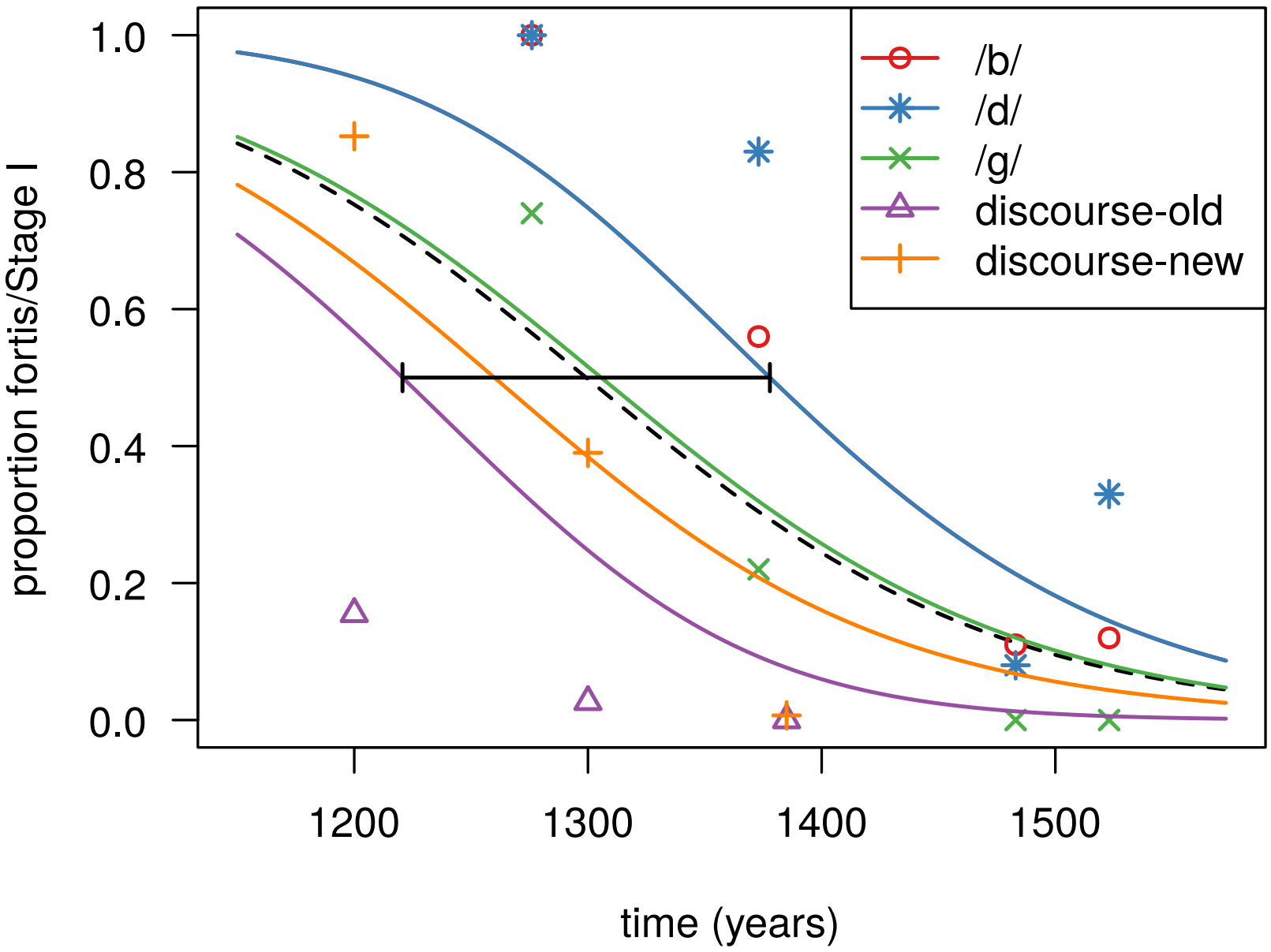
discourse-old
discourse-new

Pseudo-CRE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time (years)
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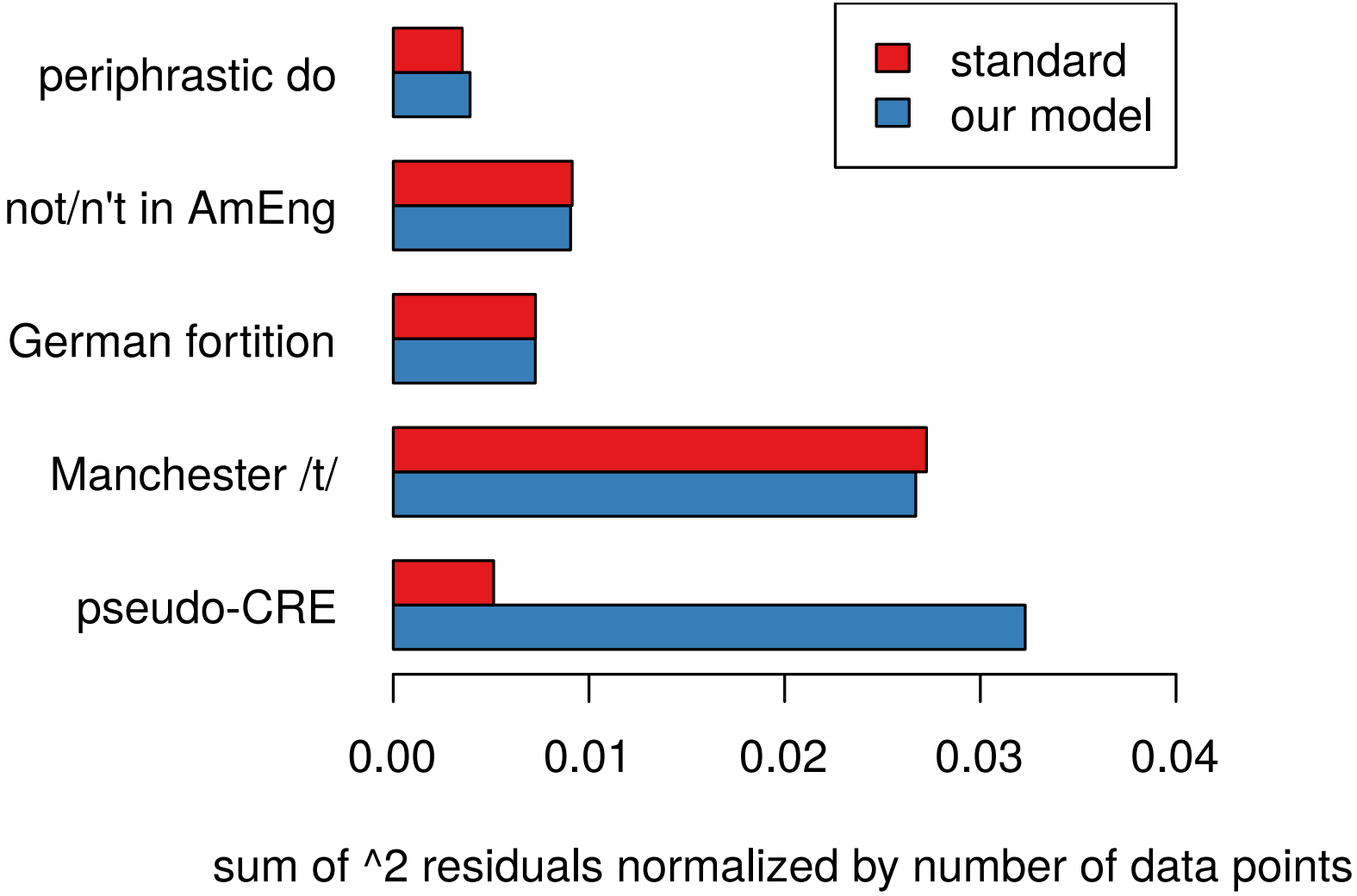
discourse-old
discourse-new

Fit


standard
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not/n't in AmEng
German fortition
Manchester /t/
pseudo-CRE
0.00
0.01
0.02
0.03
0.04
sum of ^2 residuals normalized by number of data points

4

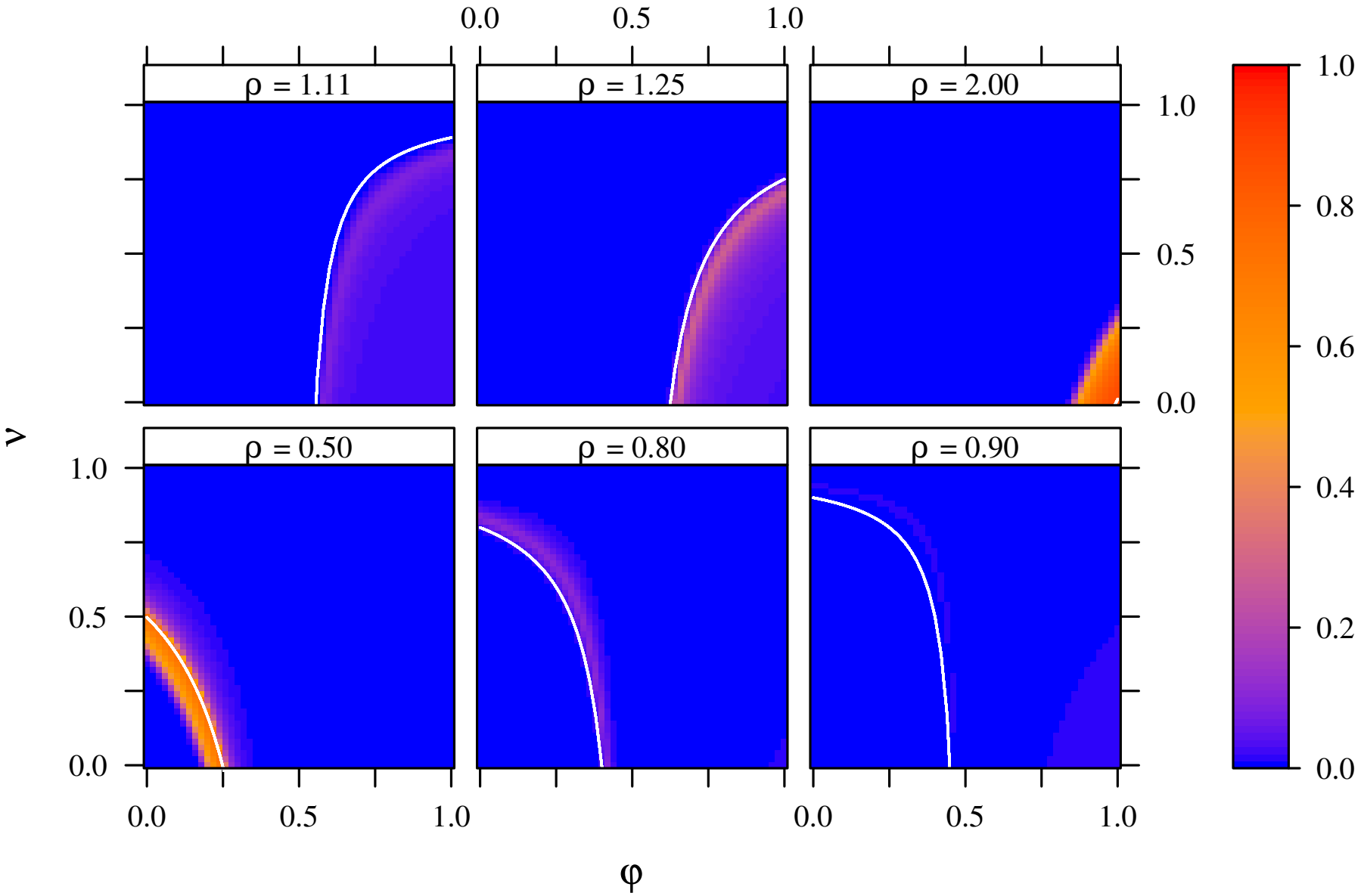
The non-logistic regime

Advantage–bias interaction

- Yang's (2000) model: logistic change, with α/β determining dynamics (The Fundamental Theorem of Language Change)
- In our model, things are more complicated:

$$q(t+1) = \left(1 + \frac{\sum_{i=1}^{M} \alpha\lambda_i(p(t) - b_i p(t)q(t))}{\sum_{i=1}^{M} \beta\lambda_i(q(t) + b_i p(t)q(t))}\right)^{-1} \tag{11}$$

- Crucially, there is a feedback loop between the underlying parametric change and the production biases
- Numerical simulations suggest that in our model underlying change is still logistic in **MOST** cases
- But not always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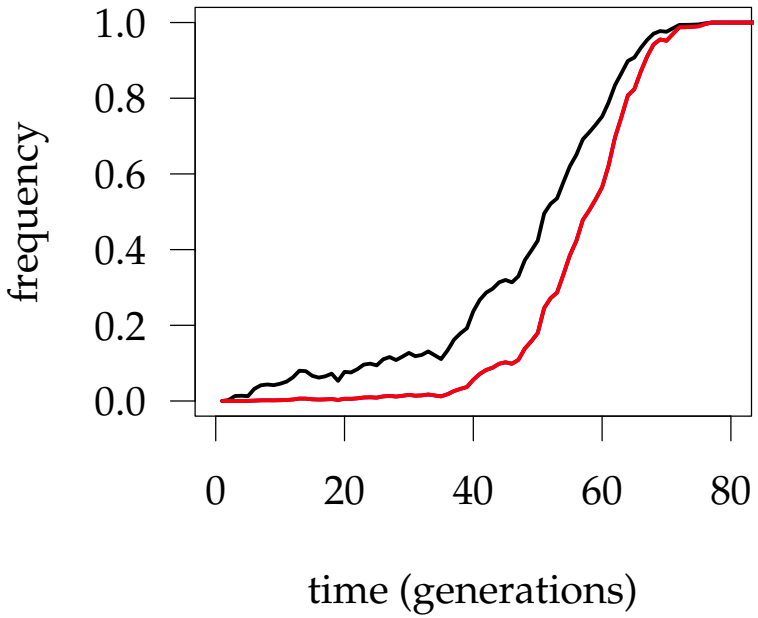
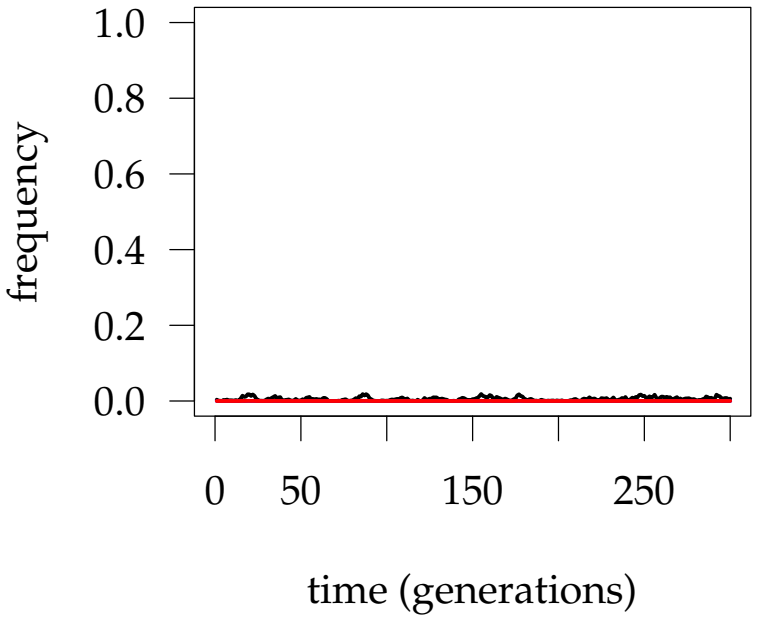
The non-logistic regime: advantage vs. bias





- Sufficiently negative production biases can
  - slow down change and make it non-logistic (still S-shaped though!)
  - block change altogether

The non-logistic regime: advantage vs. bias

Theorem (The Extended Fundamental Theorem of Language Change)

Assume reliable learners and write

$$B = \sum_{i=1}^{M} \lambda_i b_i \tag{12}$$

and

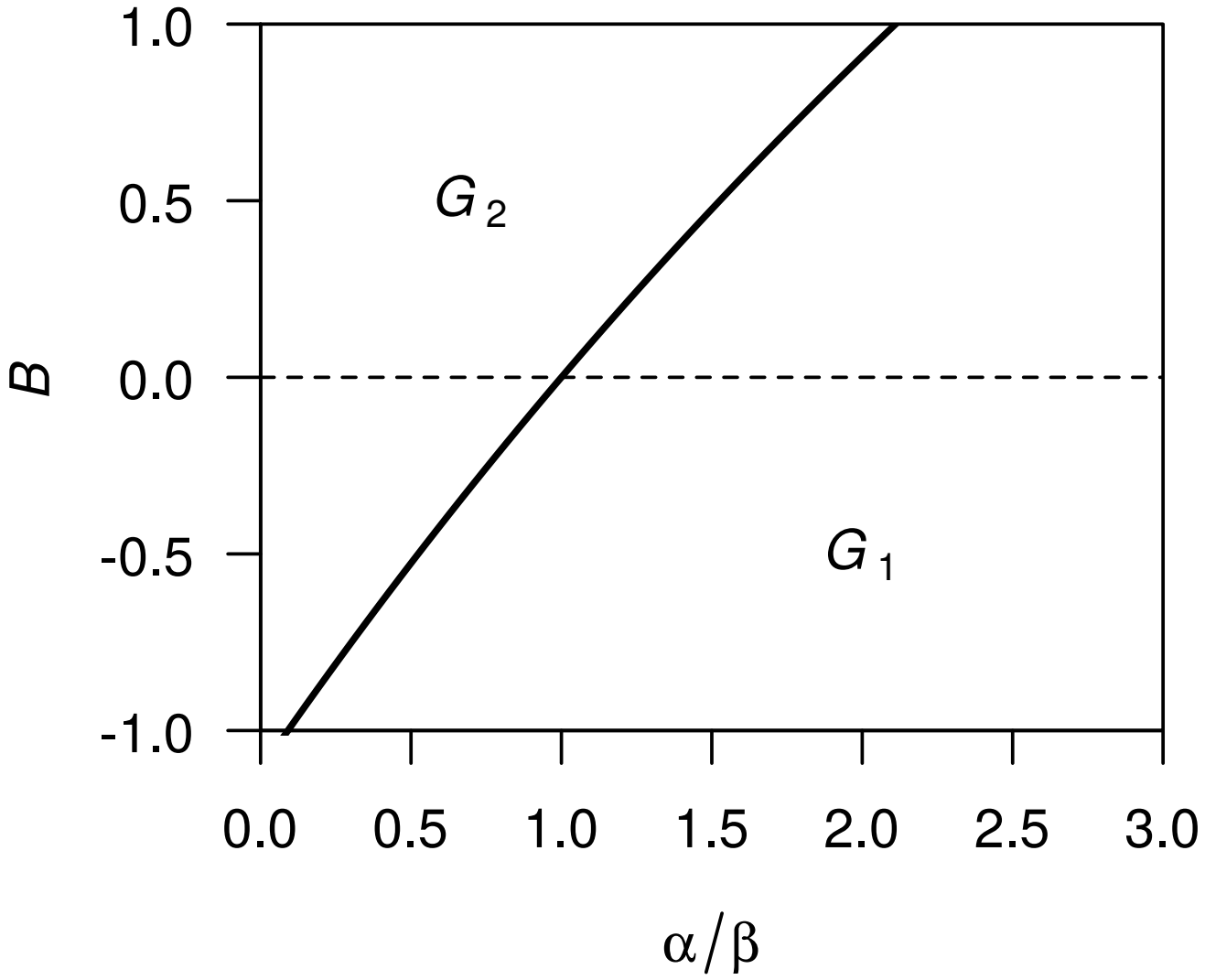
$$B_c = \frac{\alpha/\beta - 1}{1 + q(0)(\alpha/\beta - 1)}. \tag{13}$$

Then

1. $q(t) \to 1$ as $t \to \infty$, if $B > B_c$;
2. $q(t) = q(0)$ for all t, if $B = B_c$;
3. $q(t) \to 0$ as $t \to \infty$, if $B < B_c$.

In other words, G_2 overtakes G_1 if, and only if, $B > B_c$.

Bifurcation

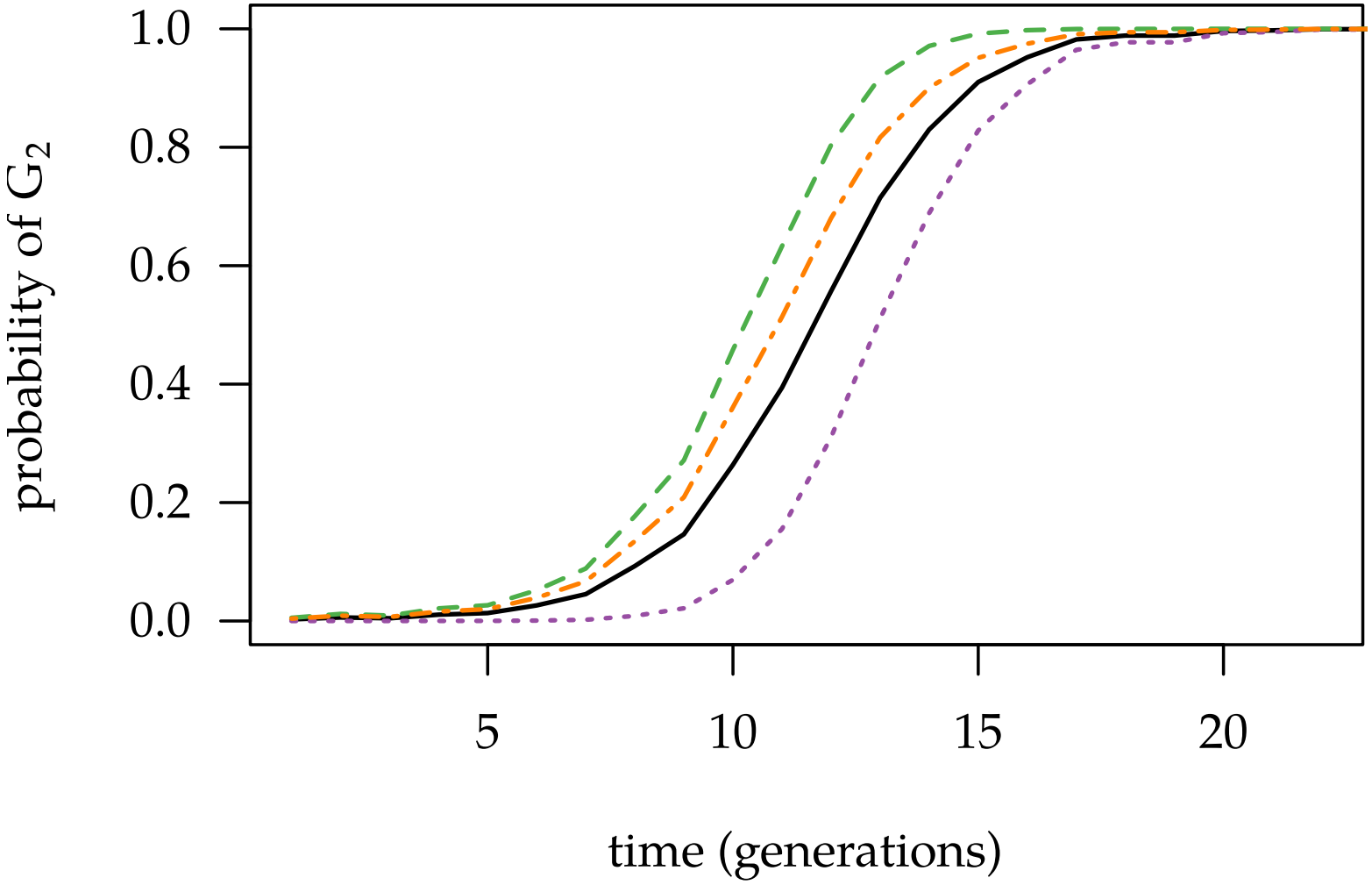


5

Conclusions

Conclusions

- We have presented a **MECHANISM** for CREs, derived from simple ingredients:
  - The variational learner of Yang (2000)
  - Iterated learning (in an infinite, well-mixing population…)
  - Production biases with fixed weights
- A corollary of this is that an upper limit can be placed on the time separation of context curves (traditionally k).
- The model gives a good fit for a number of case studies.
- Consequences:
  - Theoretical: In some instances, production biases can outweigh the parsing advantage of a grammar, so that it does not triumph in diachrony
  - Practical: Move away from the traditional "same slope, different intercepts" operationalization


1.0
0.8
0.6
0.4
0.2
0.0
probability of G_2
5
10
15
20
time (generations)

References

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